Signal Processing Simulations

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List of Notations

Notation	Description
$x(t)$	Continuous time signal
$x_s(t)$	Sampled Signal
$x[n]$	Discrete time signal
$x^{(i)}(t)$	i^{th} order derivative of $x(t)$
F_s	Sampling Rate
T_s	Sampling Period
F	Analog Frequency in Hz
Ω	Analog Angular Frequency in rad/s such that $\Omega = 2\,\pi\,F$
ω	Normalized angular frequency in rad/sample such that $-\pi \leq \omega \leq \pi$
$F_{s_{\text{ch}}}$, $T_{s_{\text{ch}}}$	Sampling rate and sampling period per channel respectively
τ	delay
α	scaling factor (can be $<$ or >1)
$\star$	convolution operation
K	Total Number of channels
$\dagger$	Pseudo Inverse Operation
$\lfloor\,\rfloor$ and $\lceil\,\rceil$	rounding down and up to the nearest integer respectively

Chapter 1

Sampling and Reconstruction

Sampling ar
Reconstruct

- *To work on some example with sinc interpolation*
- *Include simulation examples from papoulis for example*

Chapter 2

Fourier Transform

2.1 Introduction

This chapter include simulation about spectral analysis using tools such as Discrete Fourier Transfrom (DFT) and Fast Fourier Transform (FFT) .

2.2 Reminder

2.2.1 Fourier Transform Relation

Notation: taken from chapter 4 in [1]

Time: Continuous or Discrete	Analysis (Frequency Domain)	Synthesis (Time Domain)
CTFS	$c_k = \frac{1}{T_0}\int_{T_0} x(t)\, e^{-j\,k\,2\pi f_0 t} dt$	$x(t) = \sum_{k=-\infty}^{+\infty} c_k\, e^{j\,k\,2\pi F_0\, t}$
CTFT	$X(F) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi F t} dt$	$x(t) = \int_{-\infty}^{\infty} X(F)\, e^{j2\pi F t} dF$
DTFS	$c_k = \frac{1}{N}\sum_{n=0}^{N-1} x[n]\, e^{-j\,\frac{2\pi}{N}\,k\,n}$	$x[n] = \sum_{k=0}^{N-1} c_k\, e^{j\,\frac{2\pi}{N}\,k\,n}$
DTFT	$X(\omega) = \frac{1}{N}\sum_{n=-\infty}^{\infty} x[n]\, e^{-j\,\omega\,n}$	$x[n] = \int_{2\pi} X(\omega)\, e^{j\,\omega\,n} d\omega$

Table 2.1: Fourier Formula

2.2.2 Frequency Variables

- Relation between *continous frequency* F [Hz] and *continous angluar frequency* Ω [rad/s]:

$$\Omega = 2\pi F \tag{2.1}$$

- Relation between *normalized angular frequency* ω [rad/cycle] and *continuous angular frequency* Ω [rad/s] [1]

$$\omega = \frac{\Omega}{F} \tag{2.2}$$

- Moving between Hz $\leftrightarrow$ [rad/cycle]

radian/cycle $\rightarrow$ Hz	Hz $\rightarrow$ radian/cycle
$F\,[\text{Hz}] = \omega[\text{radian/cycle}] \times \left(\frac{F_s}{2\pi}\right)$	$\omega\,[\text{radian/cycle}] = F[\text{Hz}] \times \left(\frac{2\pi}{F_s}\right)$

Table 2.2: Fourier Formula

- Diagram illustration of Table 2.2[1]

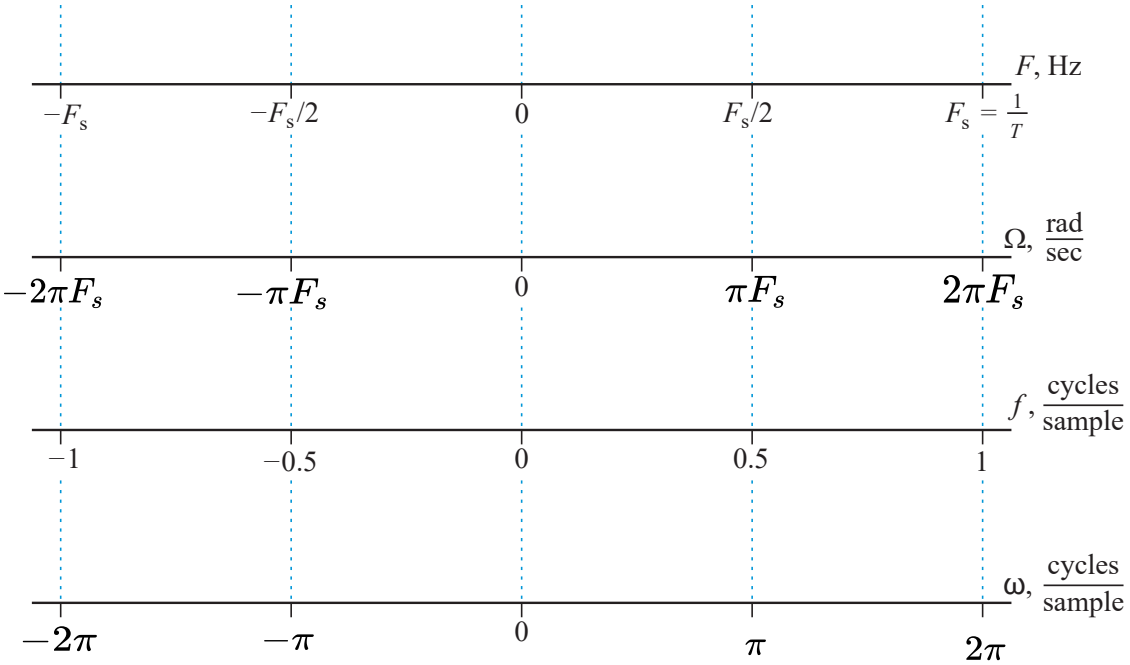


Figure 2.1: Frequency Variables illustration of Table 2.2[1]

2.2.3 DTFT-CTFT Relation

See [2].

$$\begin{aligned} X_\delta(F) &= \frac{1}{T_s}\sum_{k=-\infty}^{+\infty} X_c\,(\,F\,-\,k\,F_s\,) \\ &= X(\omega) \end{aligned} \tag{2.3}$$

- $X(\omega)$ is an DTFT of $x[n] \leftrightarrow$ DTFT of a discrete signal
- $X_c(F)$: CTFT of the $x(t) \leftrightarrow$ CTFT of a analog continous signal
- $X_\delta(F)$: CTFT of $x(n\,T_s) \leftrightarrow$ CTFT of a sampled signal

Equation 2.3 states that DTFT and CTFT of a sampled signal are the same, and if we want to retrieve the CTFT of $x(t)$, we need to multiply by T_s

DTFT-CTF Relation

: *to insert a figure for Equation 2.3.*

2.2.4 Fourier Transform Table

$x(t)$	$X(F)$
$\Pi\left(\frac{t}{\tau}\right)$	$\mathrm{sinc}(\pi\,F\,\tau)$
$2\,B\,\mathrm{sinc}(2\pi\,B\,t)$	$\Pi\left(\frac{F}{2\,B}\right)$
$\Delta\left(\frac{t}{\tau}\right)$	$\frac{\tau}{2}\,\mathrm{sinc}^2\left(\frac{\pi\,F\,\tau}{2}\right)$
$B\,\mathrm{sinc}(\pi\,B\,t)$	$\Delta\left(\frac{F}{2\,B}\right)$
$\sum_{n=-\infty}^{\infty}\delta(t\,-\,n\,T_0)$	$f_0\sum_{n=-\infty}^{\infty}\delta(F\,-\,n\,f_0)$

Table 2.3: CTFT Transform Pair

2.3 Analog Signal

Source: see example 5.8 in [2],

The continuous time domain signal to be taken is a decaying exponential:

$$x(t) = \exp(-0.5\,t)\,u(t) \leftrightarrow X(\Omega) = \frac{1}{j\,\Omega + 0.5} \tag{2.4}$$

Signal of Equation 2.4 is infinite both in time and frequency domain.
1st step is to do an approximation in frequency domain, that is choosing the maximum frequency B. Once we have B, we can apply Nyquiste theorem to have an appropriate F_s and hence a T_s, so we can truncate in time and obtain suitable number of samples.

2.3.1 Truncation in frequency domain

- Spectrum is infinite in frequency domain:

  Infinite in frequency domain means no maximum frequency to be taken to apply Nyquist sampling theorem → need to ***truncate in frequency so we can choose a sampling frequency*** F_s.

- As a criteria to truncation: what is the value of Ω at which the frequency content is bellow 2 % ?

  In other words, this Ω captures 98 % of the frequency content of the signal.

- Computations:

  - $|X(\Omega)| = \dfrac{1}{\sqrt{\Omega^2 + 0.5^2}} = 2 \times 0.01 = 0.02$
  - Solving for Ω we get: $\Omega = 50$ rad/s
  - Using Equation 2.1: $B = \dfrac{\Omega}{2\,\pi} = \dfrac{50}{2\,\pi} = \dfrac{25}{\pi}$ Hz

- The maximum frequency is $\dfrac{25}{\pi}$ Hz.

  Applying Nyquiste states that $F_{s_{\min}} = 2\,B = 2 \times \dfrac{25}{\pi} = \dfrac{50}{\pi}$ Hz

- Having $F_{s_{\min}} = \dfrac{50}{\pi}$ gives a minimum sampling period $T_{s_{\min}} = \dfrac{\pi}{50}$ s

So as summary, from frequncy truncation we get:

- $B = \dfrac{25}{\pi} \approx 8$ Hz
  - This B captures 98 % of the frequency content of the signal
- $F_{s_{\min}} = \dfrac{50}{\pi}$ Hz

2.3.2 Truncation in time domain

- Waveform in infinite in time domain:
  After we solve for $T_s = \frac{\pi}{50}$ s, there is another question to be asked: for ***how long do we need to sample*** ? $\leftrightarrow$ This signal is going forever, we can't just keep sampling forever

  So we need to solve for ***sampling duration*** $T_0 \leftrightarrow$ signal duration

- Criteria to solve the problem: We know that the signal is a decaying exponential, we can see for some value of T_0 where the ***amplitude value of the signal*** $x(t)$ ***is very low***, and by this we are ***not throwing some high value amplitude (information)***.

- Computing:
  - We try for example for $T_0 = 16$ s, we get $x(16) = e^{-0.5\,(16)} = e^{-8} = 3.3\,e^{-4}$, very small number
  - If we choose $T_0 = 16$, the number of sample we get in this period using a sampling rate $T_s = \frac{\pi}{50}$ is: $N_0 = \frac{T_0}{T_s} = 254.65$
  - However, the number of samples we need to grab need to be integer, so let's say we need to grab 256 samples
  - We resolve again for T_s using $N_0 = 256$ samples, and we get $T_s = \frac{1}{16}$ seconds

The rest is done via programming part.

2.3.3 Results

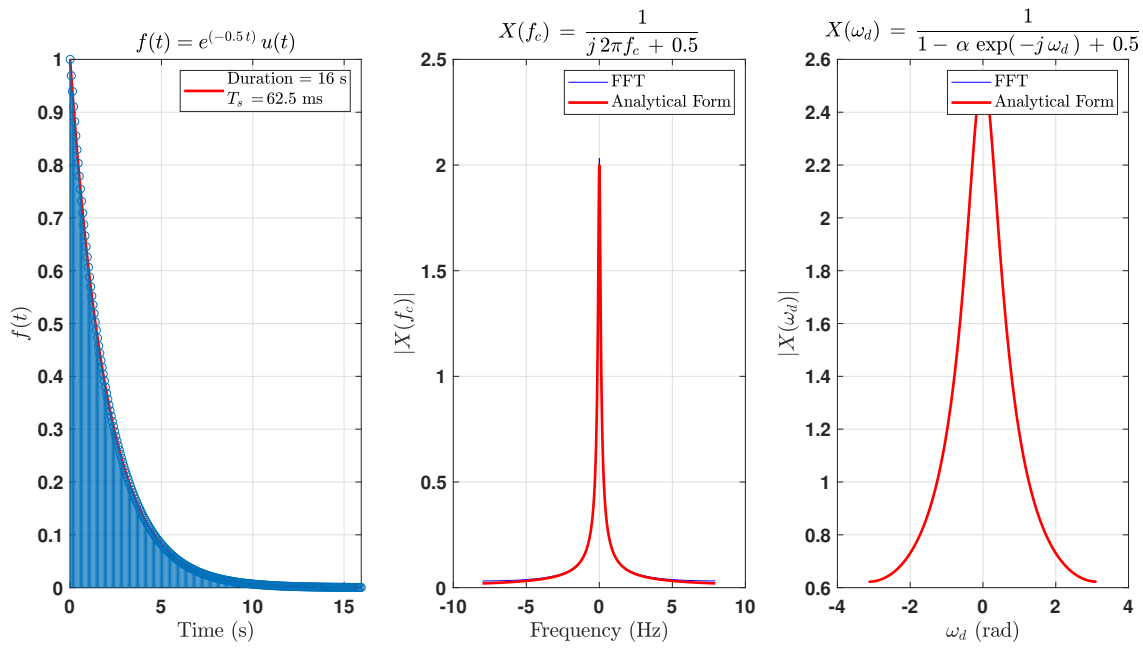


Figure 2.2: Base line: Signal Duration is 16 s and $F_s = 16$ Hz

2.3.4 Summary

As a ***summary*** we have:

- 1st question we ask since the signal $x(t)$ isn't band limited $\leftrightarrow$ keep going forever in frequency domain: ***how do we choose the sampling rate (sampling period)*** $F_s\,(T_s)$
  - what is a suitable criteria to choose some maximum frequency to represent the bandwidth B of the signal ?
  - Once we choose B, we apply Nyquiste to have a minimum sampling frequency $F_{s_{\min}}$
- 2nd question: for ***how long we keep sampling*** $\leftrightarrow$ ***(signal truncation)*** ?
  - choose some value of T_0 in which values (content) of the signal is preseverd
- Always recall that:

$$\overbrace{N}^{\text{nb of samples}} = \frac{\overbrace{T_0}^{\text{Signal duration}}}{\underbrace{T_s}_{\text{Sampling Period}}} \in \mathbb{N} \tag{2.5}$$

2.4 Working with fft

Now we turn attention to the usage of `fft` command to compute the spectrum.

In Figure 2.3, we have the result of $x(t) = B\,\mathrm{sinc}(\pi\,B\,t) \leftrightarrow \Delta\left(\frac{F}{2\,B}\right)$

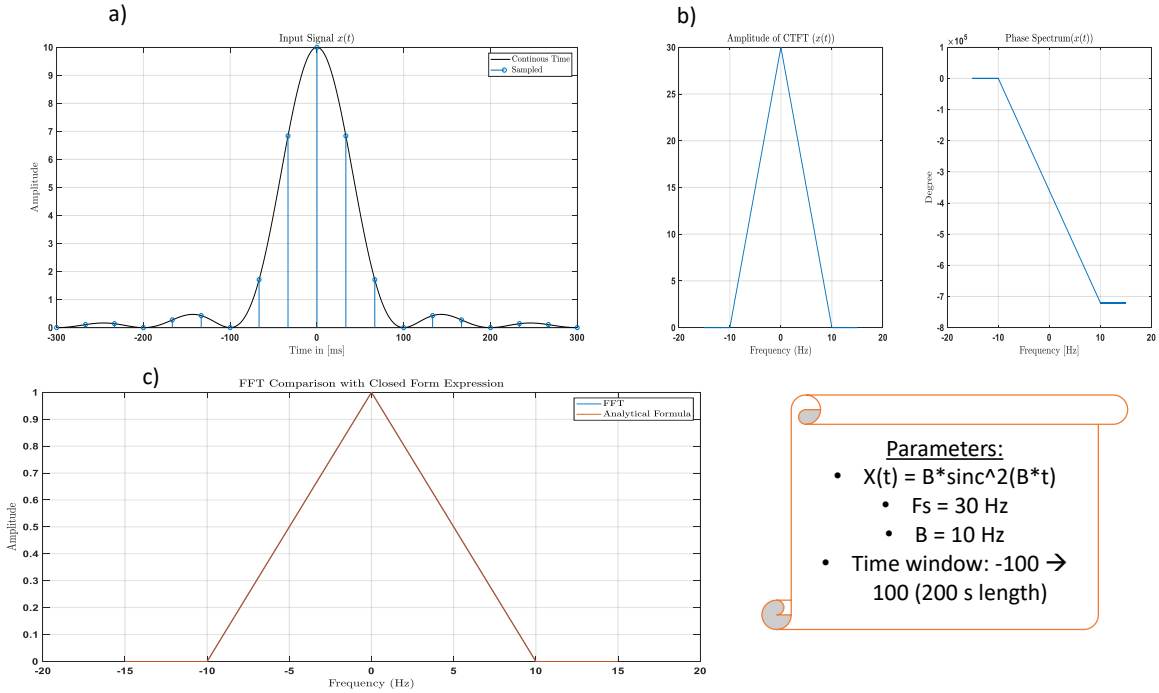


Figure 2.3: Working with sinc square

- In order to obtain the unity amplitude in $\Delta\left(\frac{F}{2\,B}\right)$, we need to divide by F_s according to Equation 2.3.

2.4.1 Rect and Sinc Simulation

Here are the result of some simulation for sinc and rect pulses presented in Figure 2.4

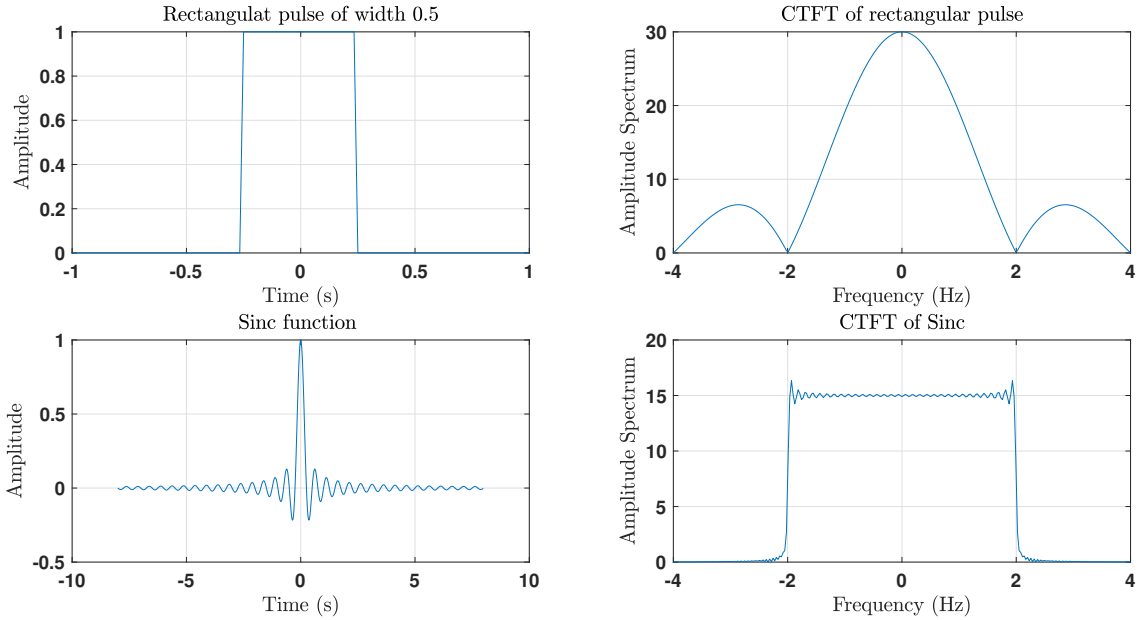


Figure 2.4: Plot of rect $\left(\frac{t}{T}\right)$ and sinc $\left(\frac{t}{T}\right)$

- T is the pulse width for the sinc and rect
  - for rect: the pulse width is divided in halfs before and after $t = 0$
  - for sinc case: pulse wdith start from $t = 0$
- Bandwidth:
  - for rect: we have a null-to-null bandwdith equal ot $\frac{1}{T} = \frac{1}{0.5} = 2$ Hz
  - Same as for sinc: however the sinc is bandlimited unlike the rect: so after 2 Hz, we have 0 values for the amplitude spectrum

2.5 IFFT

`ifft` command is used to take us back to the time domain given some discrete samples.

Note: if the input of the `ifft`,that is the complex vector (CTFT or DTFT) if shifted via `fftshift` command, we need to undo it via `ifftshift`.

Results:

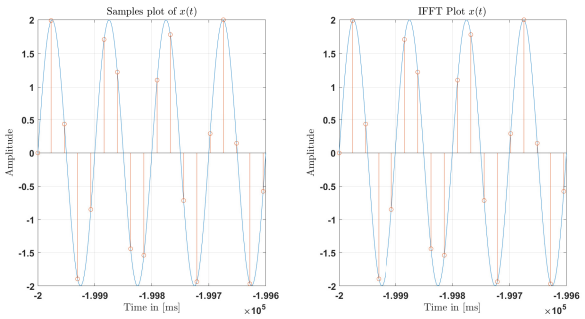


Figure 2.5: Sin wave where $f = 10$ Hz

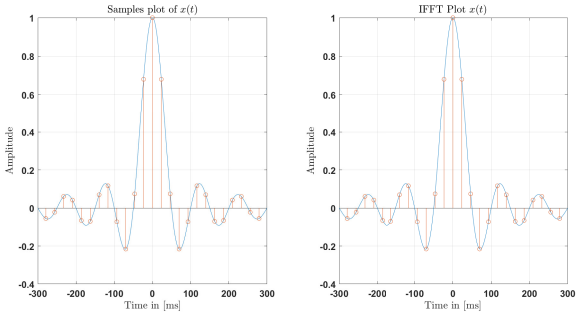


Figure 2.6: Sinc wave where $B = 10$ Hz

Even if we use zero padding in the `fft`, in the `ifft` we take a length equal to the number of samples (resulting from the uniform sampling process).

2.6 FFT over Multiple Period

IN this section, we demonstrate how to plot an FFT spectrum over multiple period.
We all know that usually we plot the FFT from $\frac{-F_s}{2} \rightarrow \frac{F_s}{2}$. However, from sampling formula we know that a sampled spectrum is period over multiple of F_s.

What we can do to implement this periodicity of spectrum is to ***set a new mapping*** between the FFT index and the true frequency bin in Hz. This mapping is governed by the relation:

$$f[\text{Hz}] = \overbrace{k}^{\text{freq index}} \times \overbrace{\frac{F_s}{N}}^{\text{freq step}} \tag{2.6}$$

such that $N =$ `length-fft`.
A normal spectrum (from $0 \rightarrow F_s$, as the mathematics of DFT says) use the following values of k:

- $k = 0 \rightarrow f[\text{Hz}] = 0$
- $k = N - 1 \rightarrow f[\text{Hz}] = \frac{N-1}{N} \times F_s$

Then to generate a spectrum centered at 0, we use the `fftshift` command, and the frequency axis using `(-N/2:N/2 - 1) x Fs/N`.

Now suppose we want a spectrum from $\frac{-3\,F_s}{2} \rightarrow \frac{3\,F_s}{2}$, the frequency bins generation will be `(-3N/2:3N/2 - 1) x Fs/N`.

An output of the simulation is shown in Figure 2.7

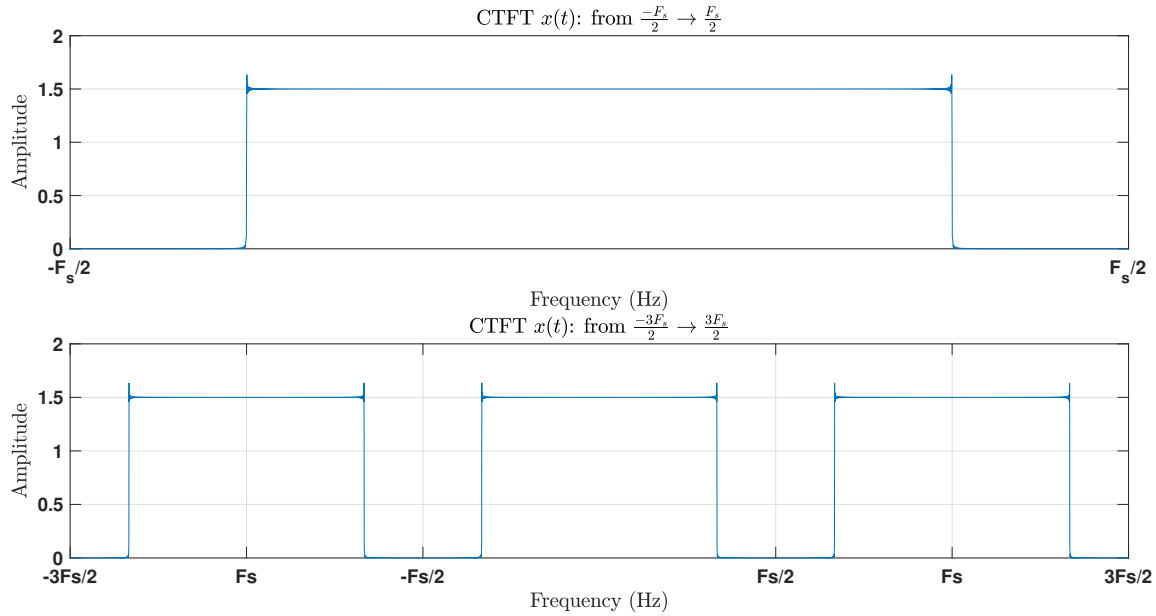


Figure 2.7: Sinc signal where $B = 10$ Hz and $F_s = 30$ Hz

2.7 Periodic Signals

In previous section the work was focused on *aperiodic continuous time signal*. Now we turn our attention to periodic signals.
We start with a square wave example, taken from [3].

- We take fundamental frequency $f_0 = 30$ Hz $\rightarrow T_0 = 0.033$s
- We set the duty cycle to 50 % $\rightarrow$ in each period, the signal is 50 % of the time up then down
- Sampling frequency is set to $F_s = 1$ KHz

Figure 2.8 shows result of simulation

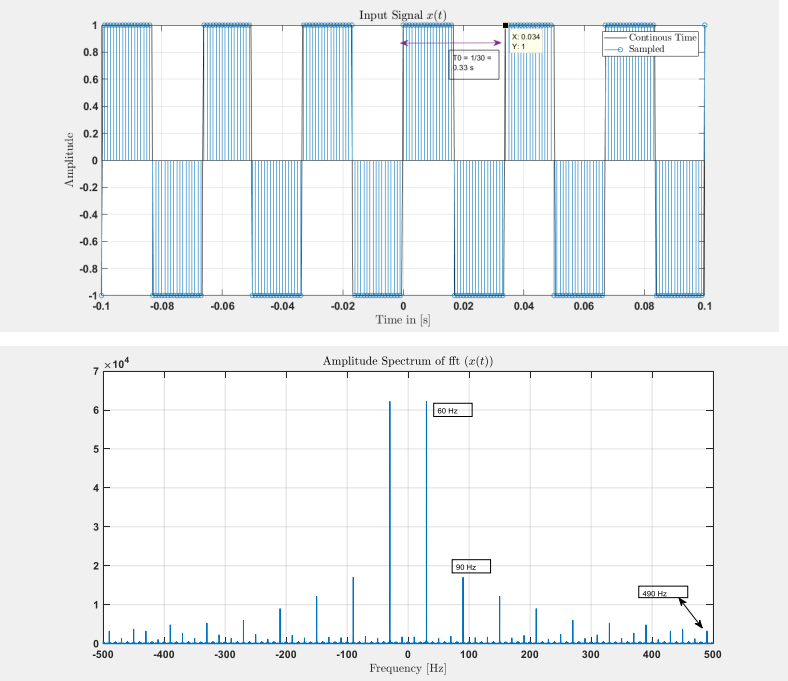


Figure 2.8: Square Wave with $F_0 = 30$ Hz

- Fourier Series expression:

$$X(t) = \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{\sin\left[(2k+1)\, 2\pi f_0\, t\right]}{(2k+1)} \tag{2.7}$$

- Harmonics appear for odd nature
  - $k = 0 \rightarrow (2k+1)f_0 = 1f_0 = 30$ Hz
  - $k = 1 \rightarrow (2k+1)f_0 = 3f_0 = 90$ Hz
  - $k = 2 \rightarrow (2k+1)f_0 = 5f_0 = 150$ Hz

2.7.1 Aliasing Effect

It is important to note that the spectrum of doesn't only contains the harmonics, but also ***aliased frequency components***.
Till now, we didn't treat the effect of aliasing in Fourier analysis, since we assume the sampling rate F_s always bigger then the maximum frequency component $f_{\max}$ in the spectrum $\rightarrow F_s > f_{\max}$. (More treatment to come about this topic in chapter 1).

Here in square wave signal, we don't have a $f_{\max}$ (notice in Equation 2.7 the frequency component keep going to ∞). Since $F_s = 1$ KHz, every frequency component above $\frac{F_s}{2} = 500$ Hz will be aliased to the range of DC $\rightarrow \frac{F_s}{2} = 500$ Hz.
As example, 510 Hz ($k = 17$) will be aliased into 490 Hz as shown in the 2nd row of Figure 2.8.

Aliasing computation: $490 = F_s - 510 = 490$

2.8 Signals used in Sampling

Source: See section 3.2.3 in [4].

- $x(t) = \delta(t) \leftrightarrow X(\Omega) = 1$
  - For a time shifted dirac: $x(t) = \delta(t - t_0) \leftrightarrow X(\Omega) = e^{j\,\Omega\,t_0}$
  - Dual transform:
    - $x(t) = 1 \leftrightarrow X(\Omega) = 2\pi\delta(\Omega)$: constant function (DC component) has its energy only at $\Omega = 0$
    - $e^{j\Omega t_0} \leftrightarrow 2\pi\,\delta(\Omega - \Omega_0)$: energy contained at $\Omega = \Omega_0$
- Train of Dirac delta function: $x(t) = \sum_{n\,\in\,\mathbb{Z}} \delta(t - n\,T_s)$

- Rectangular signal: $x(t) = \begin{cases} 1 & \frac{-1}{2} \leq t \leq \frac{1}{2} \\ 0 & \text{otherwise} \end{cases} \leftrightarrow X(\Omega) = \text{sinc}\left(\frac{\Omega}{2\pi}\right)$
  - Dual transform and using time scale property: $\text{sinc}\left(\frac{t}{T}\right) \leftrightarrow X(\Omega) = T$ for $\frac{-\pi}{T} \leq \Omega \leq \frac{\pi}{T}$

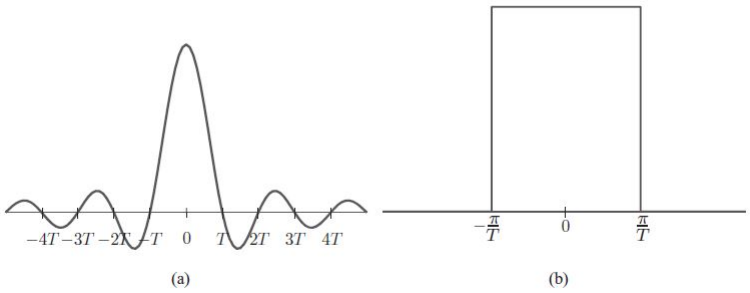


Figure 2.9: CTFT of $\text{sinc}\left(\frac{t}{T}\right)$ [4]

- Discrete Exponentials (section 3.3.2 in [4]):

$$x[n] = e^{j\,\omega_0\,n} \leftrightarrow X(\omega) = 2\pi \sum_{k \in \mathbb{Z}} \delta(\omega - \omega_0 + 2\,\pi\,k) \tag{2.8}$$

2.9 Continuous–discrete Representations

Source: See section 3.2 in [4].

In context of sampling, we deals with a ***mixed signal representation***, involving 2 components: a continuous time signal $x(t)$ and a discrete time sequence $d[n]$, having the form:

$$x(t) = \sum_{n \in \mathbb{Z}} d[n]\, h(t - n\,T_s) \tag{2.9}$$

Some facts:

- $X(\Omega) = D(\omega)\, H(\Omega)$

Chapter 3

Z-Transform

3.1 Polynomial Manipulation

Polynomial are encountered in z-transform allot.

Suppose we have the following polynomial $P(z) = 1z^0 + 1.5z^{-1} + 2z^{-2}$. We can do 3 type of manipulation:

1. Compute the root using the function `root`
2. Compute the polynomial coefficient from the root using `poly(root)`
3. Evaluating a polynomial at some value of z using function `polyval`

3.2 Partial Fraction Expansion

Example:

$$\begin{aligned} X(z) &= \frac{6 - 10\,z^{-1}\,2\,z^{-2}}{1 - 3z^{-1} + 2z^{-2}} \\ &= 1 + \frac{2}{1 - z^{-1}} + \frac{3}{1 - 2\,z^{-1}} \end{aligned} \tag{3.1}$$

In general, $X(z)$ can be written as:

$$\begin{aligned} X(z) &= \frac{B(z)}{A(z)} = \frac{b_0 + b_1\,z^{-1} + \cdots + b_M\,z^{-M}}{a_0 + a_1\,z^{-1} + \cdots + a_N\,z^{-N}} \\ &= \sum_{i=1}^{M-N} C_i\,z^{-i} + \sum_{i=1}^{N} \frac{A_i}{1 - p_i\,z^{-1}} \end{aligned} \tag{3.2}$$

If $M < N$ (proper rational function) $\rightarrow C_i = 0$.

In the example:

- poles = [1 ; 2]
- A = [2 ; 3]
- $C = 1$ since $M = N = 2$

This implemented `[terms,poles,num_constants] = residuez(coeff_num,coeff_deno)`.

3.3 Zero-Pole Pattern

Consider the system defined as

$$H(z) = \frac{1 - z^{-2}}{1 + 0.9z^{-1} + 0.6z^{-2} + 0.05z^{-3}} \tag{3.3}$$

This is a 3rd order system where $M = 2$ and $N = 3$.

To plot the zero-pole pattern in z-plane, we use the function `zplane(coeff_num,coeff_deno)`.

3.3.1 Impulse Response

We can also plot the impulse response numerically using the function `impz(coeff_num,coeff_deno,num`
If the system is stable, $h[k]$ will tend for 0 asymptotically.

3.4 Inferring Frequency Response

When evaluating frequency response from $H(z)$, we choose several value of z on the unit circle. This means that the ROC must include the unit circle so we can do this evaluation.

Usually $H(z)$ expression comes in the form of rational fraction.

1. 1st order system: $h[k] = \left(\frac{1}{3}\right)^k u[k] \mapsto H(z) = \frac{1}{1 - \frac{1}{3}z^{-1}}$, such that its ROC : $|z| > \frac{1}{3}$.
   We have a zero $= 0$ and pole at $\frac{1}{3}$.

   Recall that a rational transfer function can be written as
   $H(\omega_d) = \frac{\prod \text{Length of the Vector away from 0}}{\prod \text{Length of the Vector away from poles}}$.

   When evaluting $H(\omega_d)$ on the unit circle, the vector from 0 remains constant as we move around (because the zero it is on the origin), and the vector of the pole change.

   The pole vector will increase when we move ω_d from $0 \to \pi$, which means that the ratio will decrease. We can say that because the length of the zero vector is constant, and hence we expect a LPF behavior.

2. 2nd order system: $h[k] = \left(\frac{5}{6}\right)^k \sin\left(\frac{\pi}{3}k\right) u[k] \mapsto H(z) = \dfrac{\frac{5}{6}\sin\left(\frac{\pi}{3}\right)z^{-1}}{1 - \frac{10}{6}\cos\left(\frac{\pi}{3}\right)z^{-1} + \left(\frac{5}{6}z^{-2}\right)^2}$.

In this system we have a complex conjugate poles, having the following polar coordinates: radius $= \frac{5}{6}$ and angle $\frac{\pi}{3}$.

As we move toward high frequency components, so away from $\frac{\pi}{3}$, the distance from the poles to unit circle increase $\leftrightarrow$ influence of the poles increase $\leftrightarrow$ $|H(\omega_d)|$ decrease.

Now taking the opposite direction, by moving toward low frequency components $\leftrightarrow$ from $\frac{\pi}{3} \rightarrow 0$, the poles vector (from the poles to the unit circle) decrease, so $\leftrightarrow$ $|H(\omega_d)|$ is larger.

Influence of the radius to the Poles

If we decrease the radius, the location of the poles will push down further to the origin (further away from the unit circle), and hence we will loose the band pass effect at $\frac{\pi}{3}$.

Chapter 4

Windowing and Spectral Analysis

References: based on section 7.6 in [1] and section 4.9 in [2]

DFT (or FFT) are the computational tools which can approximate any type of Fourier transform (CTFT, CTFS,$\cdots$).
The problem of computing the DFT is always done by passing through 3 fundamental steps:

1. Sample the continuous-time signal
2. Select a finite number of samples for analysis
3. Compute the spectrum at a finite number of frequencies (frequency sampling)

Truncation is always done in real time signal processing since we have a limited amount of memory to store and process signals.

Focus in this section:effects of the ***selection of a finite-length segment of the sampled signals***, because it has the biggest impact on the accuracy of the estimated spectrum.

Assumption to be done during the analysis: the signal is sampled correctly in the time domain $\leftrightarrow$ effects of aliasing are negligible.

4.1 Modeling the problem of truncation

- Selecting a finite number of samples in the signal $\leftrightarrow$ multiplying the signal with some window
- Multiplying in the time domain $\leftrightarrow$ convolution in the frequency domain
- We consider first a rectangular window using simple cos signals, then make the analysis for other choices
- Mathematical equations:

$$x_w(t) = x(t) \times w(t) \leftrightarrow X_w(F) = X(F) \star W(F) \tag{4.1}$$

- Analysis by inspection: if we take $\Pi\left(\frac{t}{\tau}\right) \leftrightarrow \tau\,\mathrm{sinc}\,(\,\pi\,F\,\tau\,)$
  - Main lobe width $= \dfrac{4\pi}{\text{duration}}$ (see figure Figure 4.1)
  - Convolution of 2 signals: shifting and adding the 2 signals up
  - So if we have some frequency content in our signal, it will be added up (***signal will be blurred***) in the main lobe of the sinc

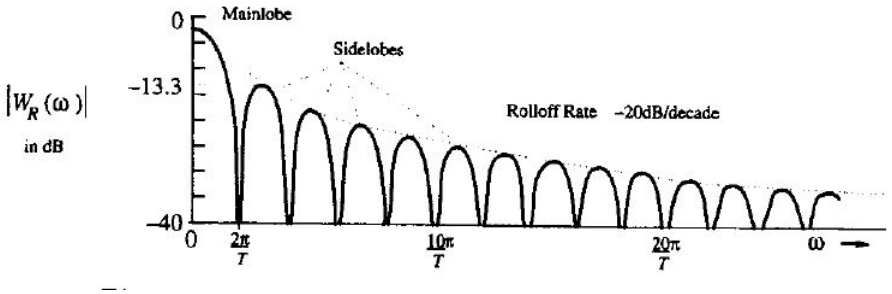


Figure 4.1: Amplitude Spectrum for Rectangular Window [2]

Example: Effect of Window with a cos Signal

- Let $x(t) = A_1\,\cos(2\pi f_0\,t + \phi)$
- The spectrum of $x(t)$ can be directly obtained using Euler identity

$$x_c(t) = \frac{1}{2}\,A_1\,e^{-j\,\phi}\,e^{-j\,2\pi f_0\,t} + \frac{1}{2}\,A_1\,e^{j\,\phi}\,e^{j\,2\pi f_0\,t} \tag{4.2}$$

- The time windowed signal $x_w(t)$:

$$x_w(t) = \frac{1}{2}\,A_1\,e^{-j\,\phi}\,w(t)\,e^{-j\,2\pi f_0\,t} + \frac{1}{2}\,A_1\,e^{j\,\phi}\,w(t)\,e^{j\,2\pi f_0\,t} \tag{4.3}$$

- Taking the CTFT of Equation 4.3 and using the frequency shift property [2] ($x(t)\,e^{j\,2\pi f_0\,t} \leftrightarrow X(F - f_0)$), we have:

$$X_w(F) = \frac{1}{2}\,A_1\,e^{-j\,\phi}\,W(\,F + f_0\,) + \frac{1}{2}\,A_1\,e^{j\,\phi}\,W(\,F - f_0\,) \tag{4.4}$$

- Interpretation of Equation 4.4: the effect of time windowing is to replace each spectral line of the CTFT spectrum with ***scaled copy of*** $W(F)$ centered at the frequency of the line
- As an example of a window, we begin with a rectangular one:

$$\Pi\left(\frac{t}{\tau}\right) \leftrightarrow \tau\,\mathrm{sinc}\,(\,\pi\,F\,\tau\,) \tag{4.5}$$

- Visualisation of the computation steps are shown in Figure 4.2

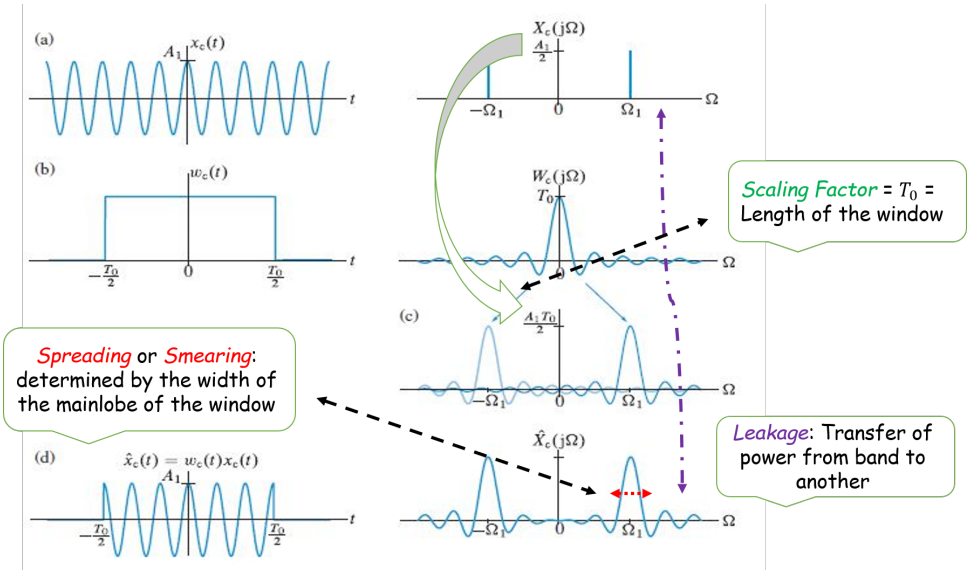


Figure 4.2: Windowing Effect on CTFT Amplitude Spectrum

3 Things to be noticed from figure Figure 4.2 [1]

1. ***Spectral Spreading*** or ***Spectral Smearing***
   - Controlled by the ***width of the main lobe***
   - The cosine signal now has spread out after and before ω_{c_1}
   - Smearing can cause ***peak merging*** $\leftrightarrow$ reduce the resolvability in the spectrum when have several components close to each other
     * To be seen in the simulations after
2. ***Leakage***:
   - Controlled by the ***sidelobs*** of the rectangular window
   - Leakage create false peaks due to the power transfer
   - Also it hides any weak frequency components as we will see in the simulation results after
3. Also notice that the amplitude spectrum is multiplied by T_0, the length of the window
   - This is because the rect window amplitude spectrum has a maximum equal to T_0

4.2 Simulation Rectangular Window

First we simulate a DTFT of a rectangular window for different length.

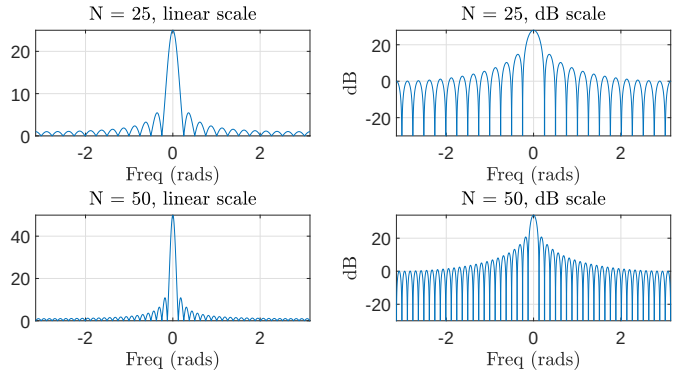


Figure 4.3: DTFT Amplitude Spectrum for the Rectangular Window in Linear and dB Scale

if we choose ***higher number of samples***

1. We get a ***narrower main lobe***
2. But the ***side lobes doesn't change that much***

4.3 Simulation 2 Cos Signals

If the ***cosine signals are too close together***, then doing convolution with our window will add up the 2 cos, and hence we can't distinguish the frequency content of the signals

- This is known of peak merging (loss of spectral resolution) when the 2 spectral line are closer then the width of the main lobe [1]

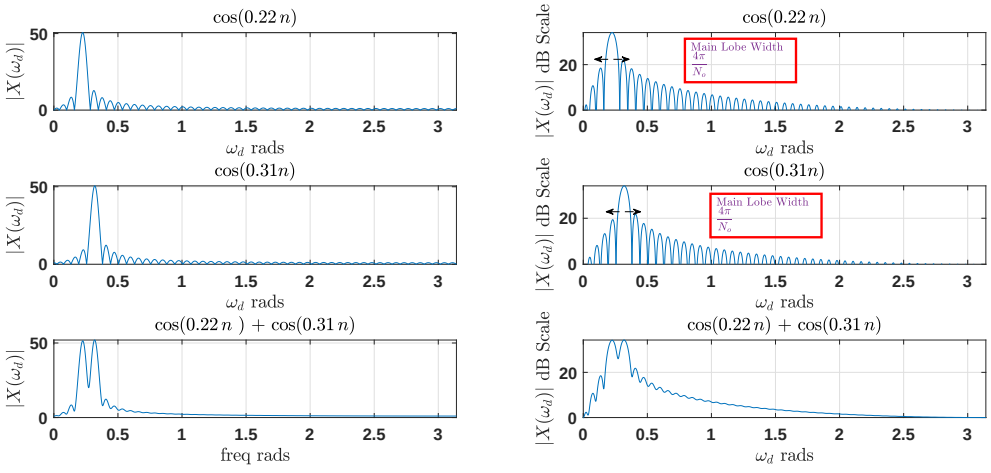


Figure 4.4: Spaced Cos signal

- We have 2 cos signals with discrete frequencies:
  - $\omega_{d_1} = \frac{7\pi}{100} \approx 0.22$ and $\omega_{d_2} = \frac{10\pi}{100} \approx 0.31$
- Width of the main lobe $= \frac{4\pi}{N} = \frac{4\pi}{100}$
  - Where N is the length of the rectangular window
- Spectral separation between signals $= \frac{10\pi}{100} - \frac{7\pi}{100} = \frac{3\pi}{100}$
- Separation rule: Width of the main lobe $\leq$ Spectral separation between signals $\leftrightarrow \frac{4\pi}{100} \leq \frac{3\pi}{100} \leftrightarrow$ rule is almost verified with some errors however
  - But still we are able to distinguish the 2 components

1 Weak and 1 Strong Signal

Trickiest part: don't forget ***the side lobes***

- Case of 1 strong and the 2nd is weak
- Even if the 2nd component is far from the 1st one, the 2nd could be a small one and add up with the side lobes of the 1st one (the strong one)

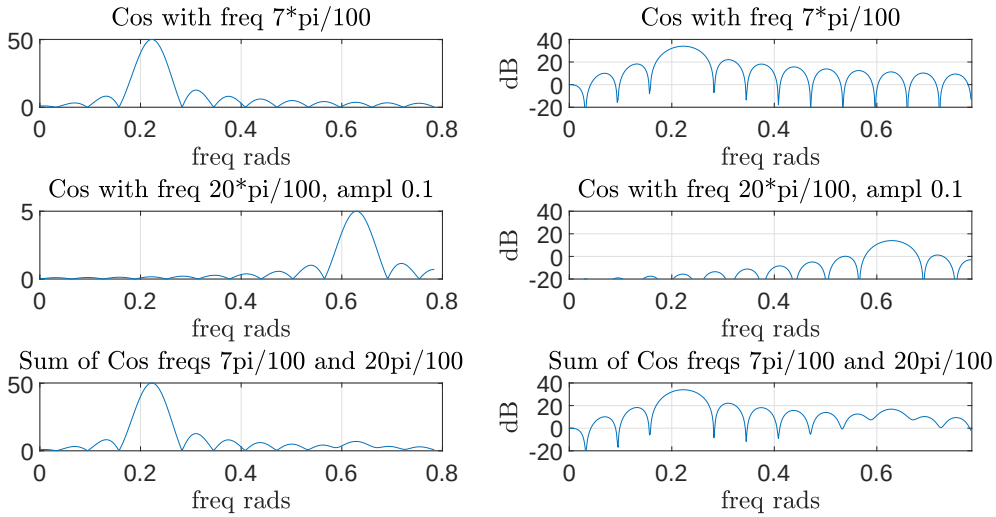


Figure 4.5: Well separated signals but with weak components

4.4 Window Choice

- In previous part, we have seen the effect of side lobs in a rectangular window:
  1. Very high resolution ↔ narrow main lobe
  2. Not very efficient dynamic range ↔ side lobe effects still high even if we truncate less
- In rectangular window, we have a sudden transition from $1 \to 0$ or $0 \to 1$ in the time domain, which make the side lobes to be appear in the frequency domain (in CTFT or DTFT Spectrum)
- This was the motivation for designing other choices of windows: ***trade-off between resolution and dynamic range***
- Design criteria for ***good window***:
  1. Narrow mainlobe
  2. Weak side lobes
- Different Window Types

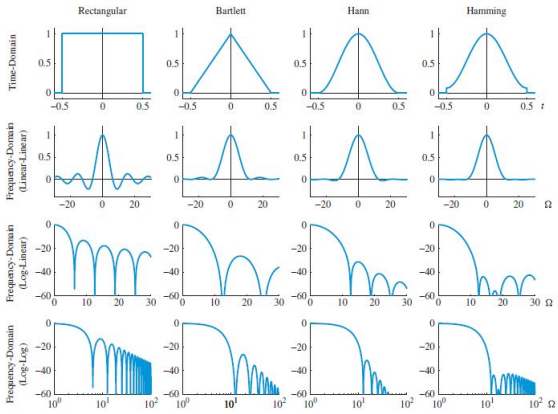


Figure 4.6: Time-domain and frequency-domain characteristics of continuous-time rectangular, Bartlett (triangular), Hann, and Hamming windows [1]

Window	Mainlobe width	Rolloff rate (dB/octave)	Peak sidelobe level (dB)
Rectangular	$4\pi/T_0$	−6	−13.3
Bartlett	$8\pi/T_0$	−12	−26.5
Hann	$8\pi/T_0$	−18	−31.5
Hamming	$8\pi/T_0$	−6	−42.9

Figure 4.7: Some continuous-time windows and their characteristics [1]

- Rectangular window has the narrower main lobe ↔ highest resolution
  - However, their side lobes are relatively high ↔ poorest dynamic range

- Hamming window has a smoother transition in the time domain from $1 \rightarrow 0$
  - Hence it has a wider mainlobe compared to the rectangular window $\leftrightarrow$ lower resolution
  - But lower side lobes $\leftrightarrow$ higher dynamic range
- Chebyscheve window: we can control the side lobes height, and it has equal height side lobes
  - The weaker the side lobes the more the transition is smooth
  - Chebychev with -30 dB attenuation: narrower main lobe but high side lobes
  - Chebychev with -70 db has quite the opposite compared to the -30 dB attenuation
- As a general rule: the more we push down the side lobes in the frequency domain (hence better dynamic range) $\leftrightarrow$ the main lobe will get higher (lower resolution)

4.5 DTFT Spectrum Using Different Window Type

- Example: $x[k] = \sum_{i=1}^{i=3} \text{Components}_i$
  - Component 1 $= \cos\left(\frac{\pi}{10}k\right)$
  - Component 2 $= \cos\left(\frac{\pi}{8}k + \frac{\pi}{3}\right)$
  - Component 3 $= 0.01\cos\left(\frac{4\pi}{10}k - \frac{3\pi}{5}\right)$
- 2 criteria to be tested via the discrete signal $x[k]$: Resolution and Dynamic Range
  - Resolution: the sum of 2 close signal $\leftrightarrow$ controlled by the width the main lobe
  - Dynamic Range: displaying weak component $\leftrightarrow$ controlled by the height of the side lobes

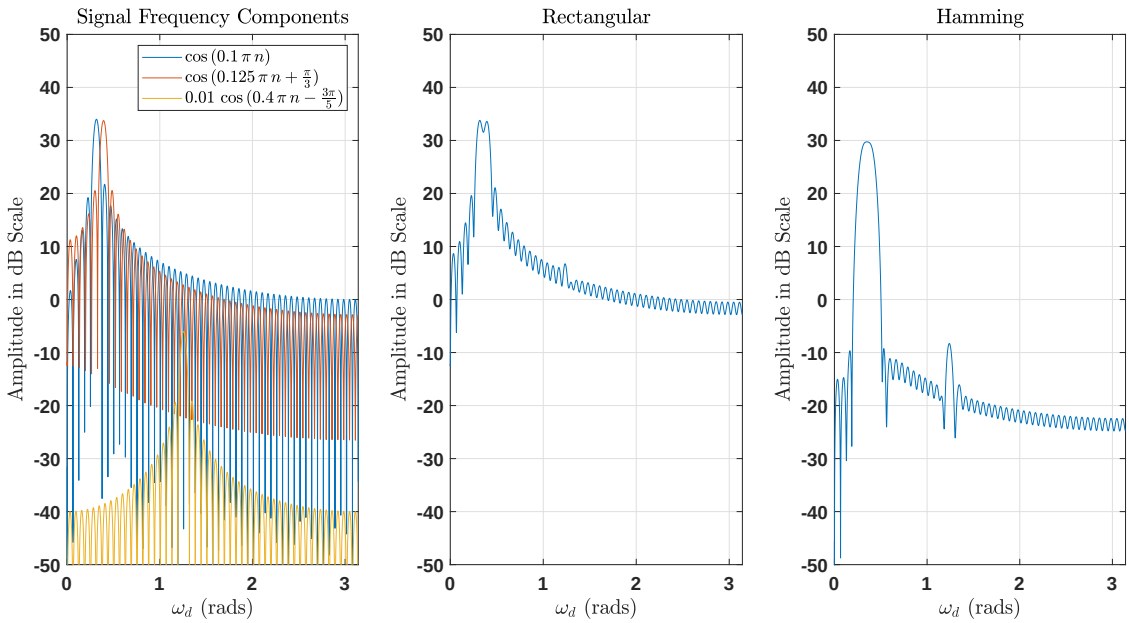


Figure 4.8: Rectangular and Hamming Window

- In rectangular window:
  - the 2 components can be distinguished
  - However the 3rd components having a small amplitude has disappeared in the side lobes
    - It barely we can see it
    - If we have some noisy data, it will completely gone
- In Hamming window:
  - The 2 cos peaks has merged
  - But the weak components is detected better then the rectangular window

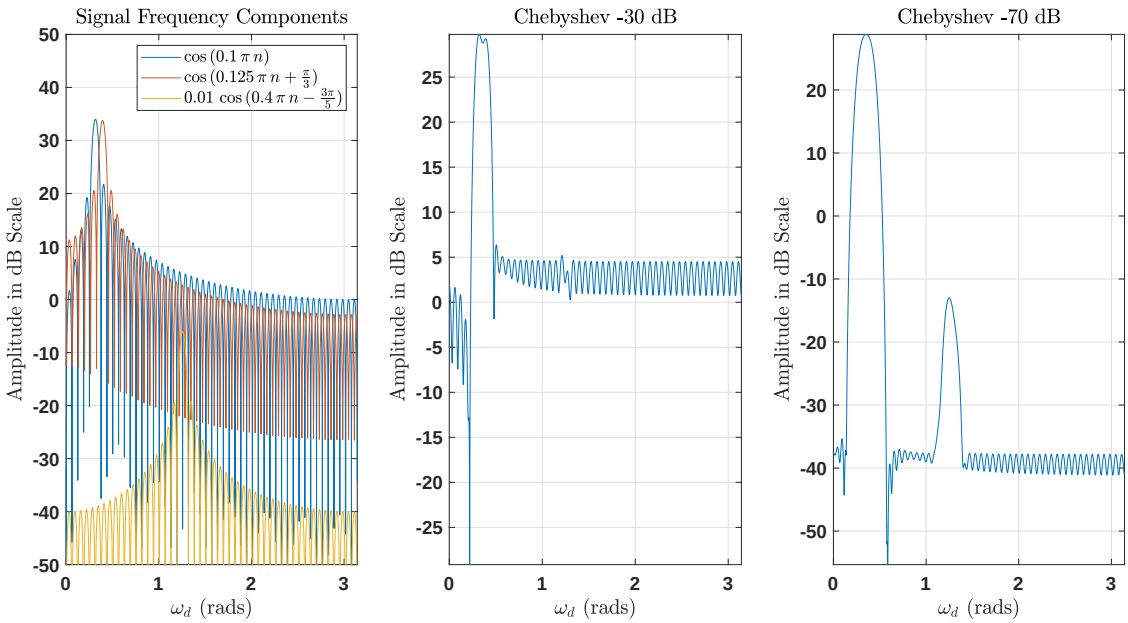


Figure 4.9: Chebychves Window with -30 dB and -70 dB side lobes attenuation

- In -30 dB case:
  - We can (although it is difficult) to see that we have 2 frequency components $\leftrightarrow$ good resolution
  - However the weak component have vanished $\leftrightarrow$ poorer dynamic range
- In -70 dB case we have opposite results
  - Since thee side lobes now are more attenuated, we can detect the weak component
  - Of course in the price of poorer resolution as we can see, the 2 components are gone now

4.6 CTFT Analysis Using Different Window

- We try to see now the effect of windows on CTFT spectrum for a damped cos signals given by:

$$x(t) = e^{-0.1\,t}\,[\cos(10\,t) + \cos(12\,t)]\,u(t)$$

$$X(\omega_c) = \frac{0.1 + j\,\omega_c}{(0.1 + j\,\omega_c)^2 + 10^2} + \frac{0.1 + j\,\omega_c}{(0.1 + j\,\omega_c)^2 + 12^2}$$

- $X(\omega_c)$ is not a bandlimited signal, so we need to choose a maximum frequency
  - Since we can visualize that after 12 rad/s the spectrum roll off, we choose effective bandwidth of $B = 300$ rad/s $\rightarrow T_s = 0.01$ s

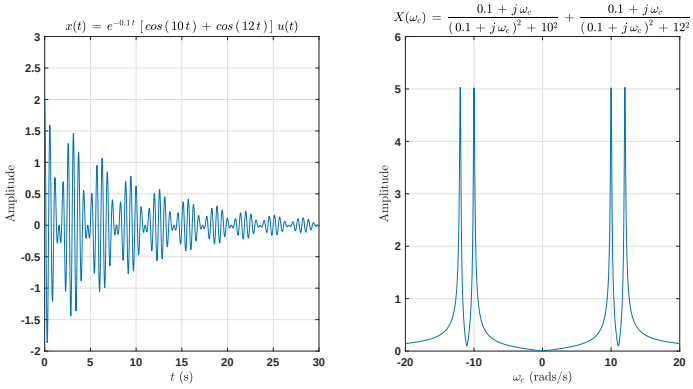
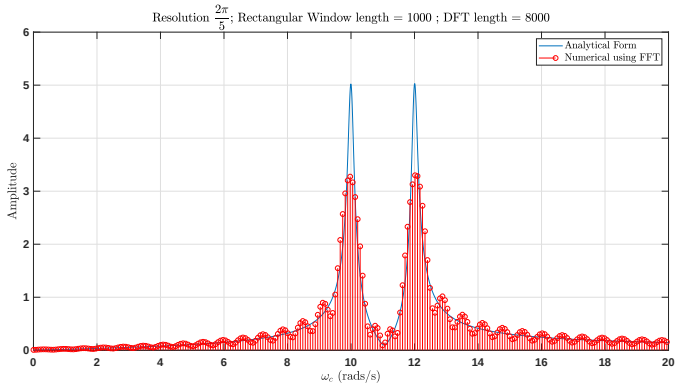
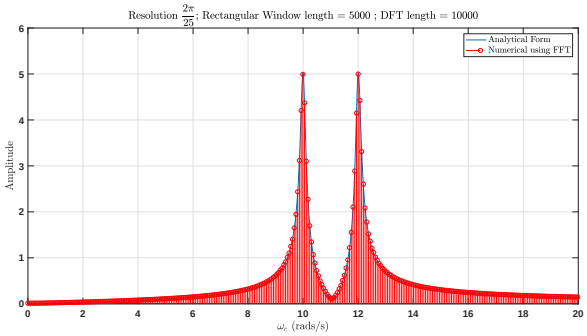


Figure 4.10: $x(t)$ along with its Amplitude Spectrum

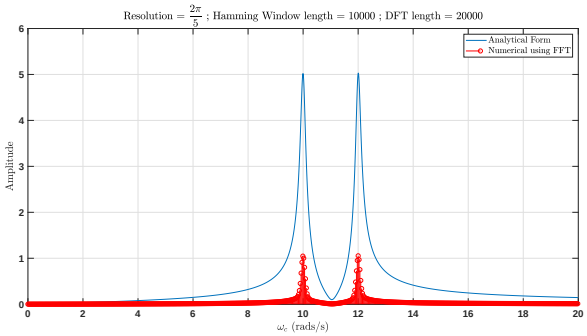
- We have 2 damped frequency components at 10 and 12 rad/s. Spectral Separation between these 2 components is $12 - 10 = 2$ rad/s
- Recall that the width of the main lobe need to be smaller than the spectral separation
- We choose to see details at 2 resolution using rectangular window:
  1. $\frac{2\pi}{5} \approx 1.2566 << 2$ rad/s
     - $\frac{4\pi}{\text{Signal Duration}} = \frac{4\pi}{T_s\,N} \leq \frac{2\pi}{5} \rightarrow N \geq 1000$ samples
  2. We increase resolution times 5: $\frac{2\pi}{25} \approx 0.2513$
     - $\frac{4\pi}{\text{Signal Duration}} = \frac{4\pi}{T_s\,N} \leq \frac{2\pi}{25} \rightarrow N \geq 5000$ samples



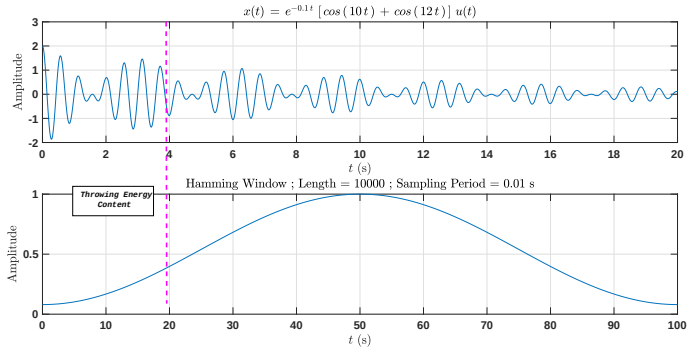
  - DFT Values are distorted
  - Some ripples at the tails
- Increasing resolution 5 times from $\frac{2\pi}{5} \rightarrow \frac{2\pi}{25}$ (narrower main lobe)
  - Better DFT Values and No ripples at the tails



- Trying Different type of Windows: Hamming window for example, with the best observed resolution which is $\frac{2\pi}{25}$
- Hamming Length:
  - $\frac{8\pi}{\text{Signal Duration}} = \frac{8\pi}{T_s N} \leq \frac{2\pi}{5} \rightarrow N \geq 10^4$ samples
  - Rectangular window length was 5000 samples for a resolution of $\frac{2\pi}{25}$
  - Hamming need 2 times more samples then rectangular to observe same level of details



- DFT amplitude values are very distorted
- Hamming is attenuating signal energy when we are windowing



Chapter 5

Complex Number and Low Pass Simulation

References: chapter 5 in [5] and chapter 4 in [6]

5.1 Review of Basics

- Advantage of complex notation: easier to work with multiplication of complex exponential rather then the multiplication of 2 cos
  - $e^{j\,A}\,e^{j\,B} = e^{j\,(A+B)}$
  - $\cos(A)\,\cos(B) = \frac{1}{2}\,[\cos(A + B) + \cos(A - B)]$
- Signal processing work with voltages to process real world quantities
  - When the antenna see the change in the EM field, it induces voltages
  - Microphone pick up sound waves from small changes in sound pressure to produce voltage

5.2 Quadrature Representation

5.2.1 Case Usage

IQ signals are used in many places. In communication applications:

- Used in Tx side to ***unify many modulation scheme*** (as shown in Figure 5.3)
  - Instead of using different circuitry for each modulation type, we can change the IQ values to obtain various modulation type
- Used in Rx for demodulation as shown in Figure 5.1

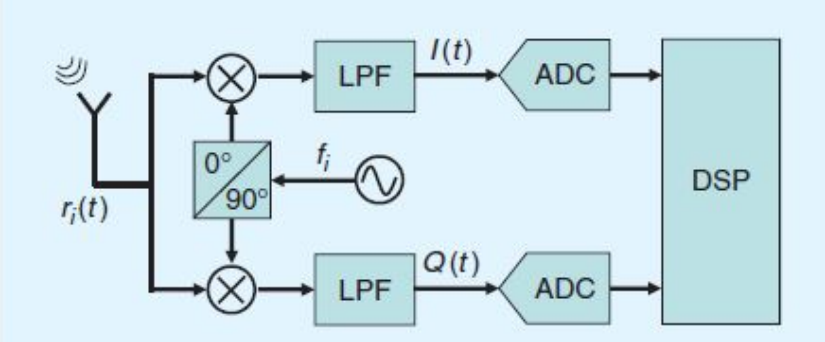


Figure 5.1: A block diagram of a typical I-Q demodulator [7]

e Usage

Try to simulate these idea in practical application

5.2.2 Formula

$$\begin{aligned} x_{\text{band-pass}}(t) &= \text{Amp}(t) \times \cos\left(2\pi\, f_{\text{center}}\, t + \text{phase}(t)\right) \\ &= x_{\text{in-phase}}(t)\cos\left(2\pi\, f_{\text{center}}\, t\right) + x_{\text{quadrature}}(t)\sin\left(2\pi\, f_{\text{center}}\, t\right) \end{aligned} \tag{5.1}$$

where
$$\begin{cases} x_{\text{Eq-Baseband}}(t) = \text{Amp}(t) \times \exp\left[j \times \text{phase}(t)\right] \\ \text{Amp}(t) = \sqrt{x^2_{\text{in-phase}}(t) + x^2_{\text{quadrature}}(t)} \\ \text{phase}(t) = \tan^{-1}\left[\dfrac{x_{\text{quadrature}}(t)}{x_{\text{in-phase}}(t)}\right] \end{cases}$$

5.2.3 Summary for IQ Representation

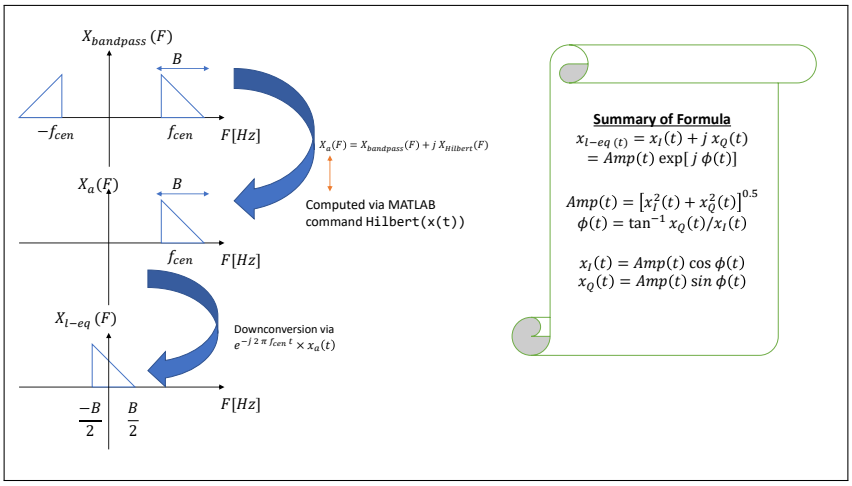


Figure 5.2: Summary for IQ Representation

5.2.4 Analog Generation of IQ Signals

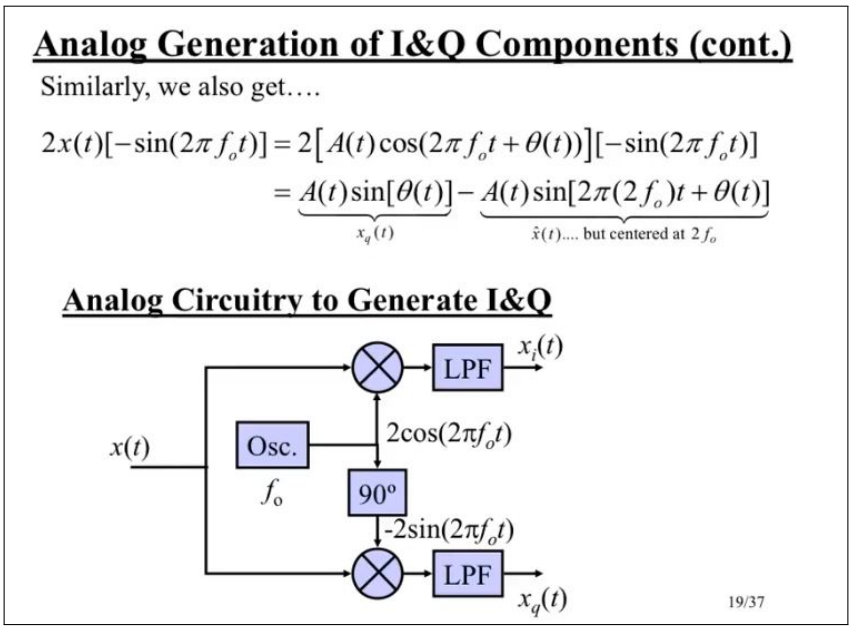


Figure 5.3: IQ generation and implementation using analog devices

5.3 Examples

FT leakage

- Ex 5.1 page 175 in [5]: contains nice explanation about leakage and bins resolution

5.3.1 AM Signals

Let the signal $x(t)$ given by:

$$x(t) = \tag{5.2}$$

5.3.2 FM Signals

First we recall the FM signal model:

$$x_{\mathrm{FM}}(t) = A_c \cos\left(2\pi f_c t + 2\pi K_{\mathrm{FM}} \times \int_{-\infty}^{t} m(t)\,dt\right) \tag{5.3}$$

In case of FM, $\Phi(t) = \int_{-\infty}^{t} m(t)\,dt$, hence the IQ components are:

$$I_{\mathrm{FM}}(t) = A_c \cos\left(2\pi K_{\mathrm{FM}} \int_{-\infty}^{t} m(t)\,dt\right) \tag{5.4}$$

$$\tag{5.5}$$

$$Q_{\mathrm{FM}}(t) = A_c \sin\left(2\pi K_{\mathrm{FM}} \int_{-\infty}^{t} m(t)\,dt\right) \tag{5.6}$$

5.3.3 Design Criteria: Carrier Synchronization and Phase locked Loop

Source:

- Begin with a refresher about AM-DSB-SC, section 5 page 182 [5]
- Phase Error See section 5.3.3 in [5] :
  - In demodulation, the oscillator may not be synchronized with the carrier $\leftrightarrow$ we have some phase θ
  - For a base band message $g(t) = A\cos(2\pi f_b t)$, the demodulation will give: $\hat{m}(t) = \frac{A}{2} g(t)\cos(\theta)$
  - for $\theta = \pm\frac{\pi}{2}$, $\hat{m}(t) = 0$, that's why we need carrier synchronization and phase locking at the Rx side
- QAM: a method to transmit 2 data streams on the same carrier, but with a separate offset

  This a more bandwidth efficient scheme.

  If we want to compare it to AM-DSB-SC: we need 2 times the bandwidth in AM-DSB-SC for transmission, but in QAM we only need f_b.
  - QAM Tx Side

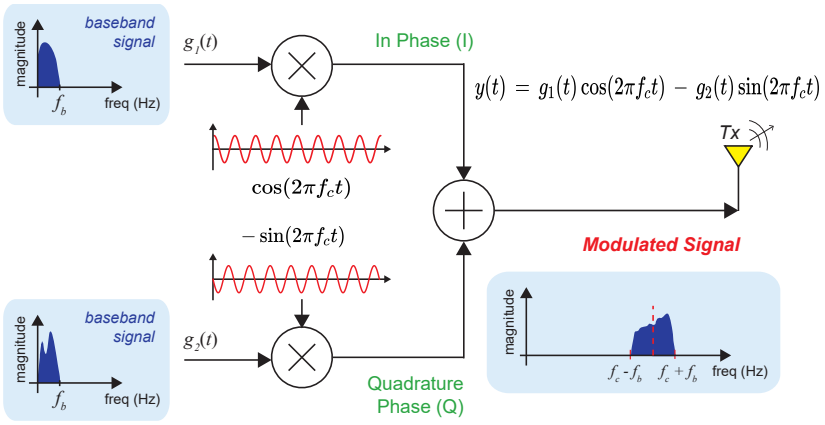


Figure 5.4: QAM Tx Side

  - QAM Rx Side

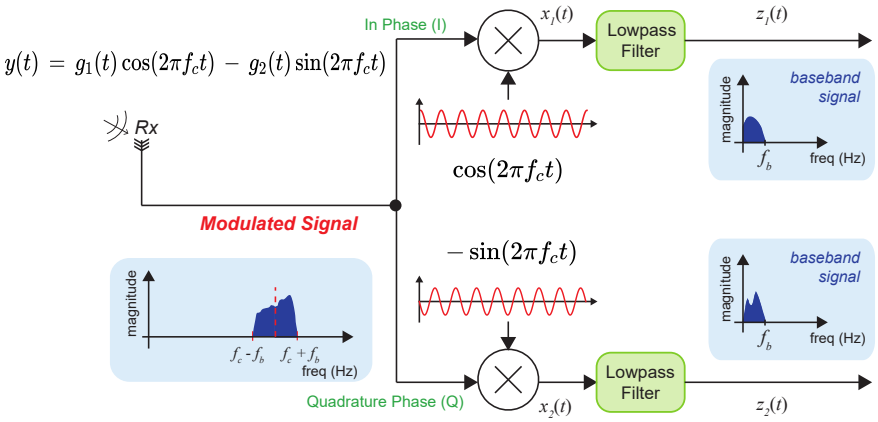


Figure 5.5: QAM Rx Side

- Using Complex Notation: the key in complex notation is to remember Equation 5.7

$$\cos(2\pi f_c t) = \text{Re}(e^{j2\pi f_c t}) \tag{5.7}$$

In Figure 5.6, we have the equivalent Tx-Rx QAM scheme using complex notation

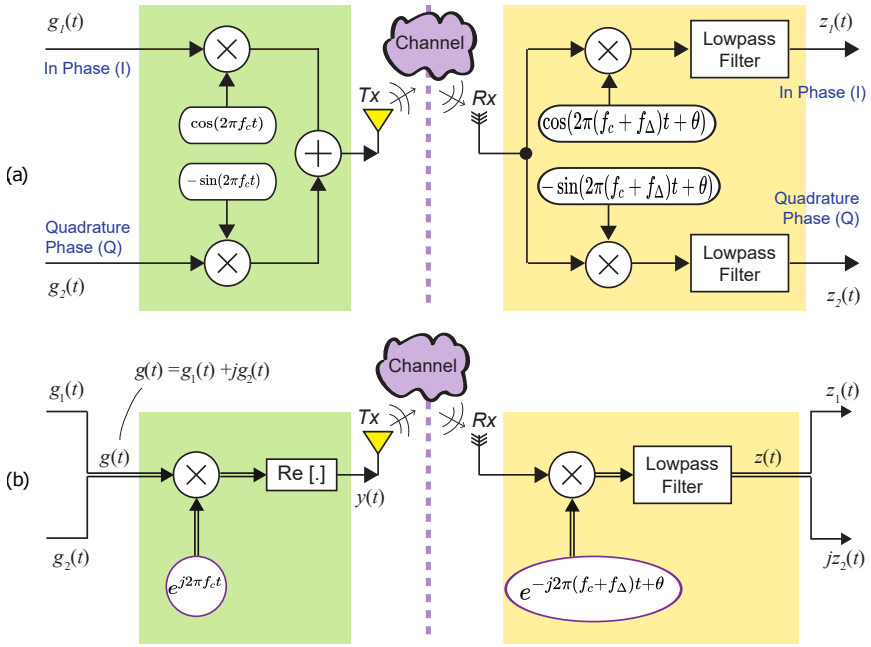


Figure 5.6: QAM Tx-Rx Using Complex Notation

Chapter 6

Power Spectral Density

6.1 Introduction

Spectral Analysis: characterize power distribution over frequency using finite amount of time series data (chapter 1 in [8]).

The PSD concept is important to use in communication signals, where they are random in nature, and hence they have infinite power. To analyze such signals, PSD concept is used (see section 1.2.4 [9])
For a nice introduction and reviewing concept:

- See chapter 1 (mainly 1.3) in [8]

6.2 Pwelsh Method

We start by simple simulation of a sin function:

$$x(t) = A \sin 2\pi f t \quad \text{where } f = 500\text{Hz and } A = \sqrt{2} \tag{6.1}$$

We take $F_s = 4$KHz and number of point to generate 1024.

Analytically, power of $x(t) = \dfrac{A^2}{2} = 1$ Watt $\rightarrow 10\log_{10}(1) = 0$ dBW/Hz.

6.2.1 Reminder of Power Notion

See section 2.5 in [10].

- PSD gives power per frequency components $\leftrightarrow$ Watt/Hz
- The ***total power*** or ***average power*** is the sum of the PSD (integration to be precise)
- For sin and cos signals:
  - PSD is $\dfrac{A^2}{4}$
  - Average power $= \dfrac{A^2}{4} + \dfrac{A^2}{4} = \dfrac{A^2}{2}$

6.2.2 Matlab Simulation

In Maltab, we can use the `pwelsh` function to compute the average power where: `[pxx,f]= pwelch(signal,window,n_overlapp,nfft,fs);` where:

- `nfft = Nb_samples =` 1024
- `window = rectwin(nfft);`

Figure 6.1 show the results of the simulation.

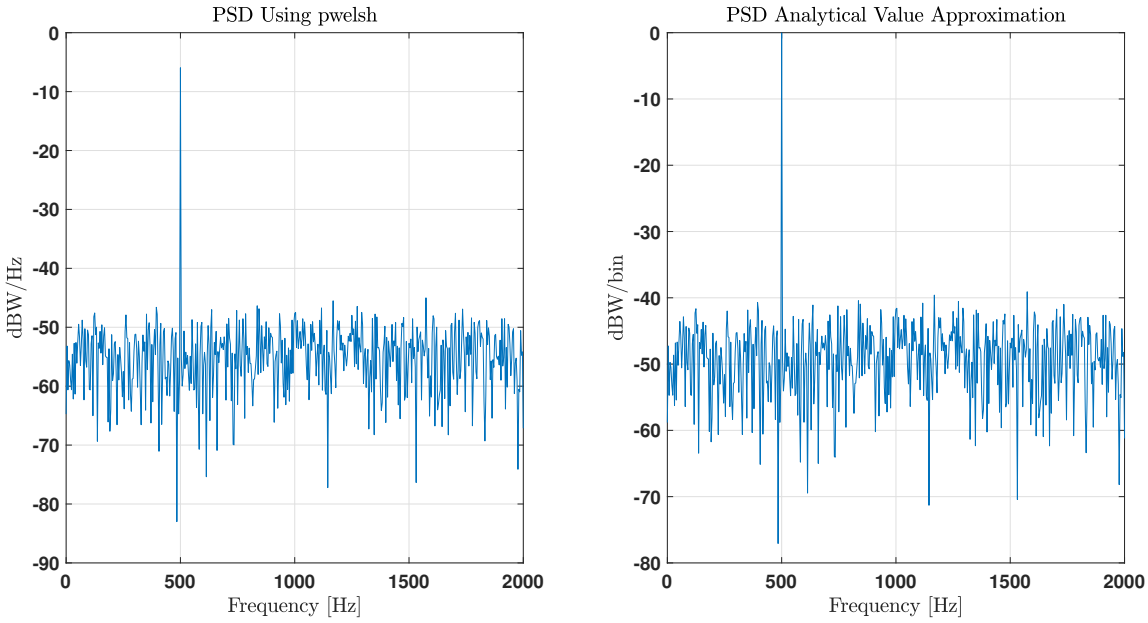


Figure 6.1: Sin Demo using pwelsh function

The result of the `pwelsh`, which is `10log10(pxx)`, give a value of ≈ -5.9 dBW, which is different from 0 dBW.

So how to move from -5.9 dBW → 0 dBW ?
What we need to do is to multiply the value of `pxx`, which is W/Hz, and convert to Watt/bin, and this done as:

$$\text{Pbin (W/bin)} = \text{pxx (W/Hz)} * \text{Fs/nfft (Hz/bin)} \tag{6.2}$$

6.2.3 Windowing,Overlapping

D:Windowing,Overlapping

.

6.3 Equivalent noise bandwidth(ENBW)

- source: https://analog.intgckts.com/equivalent-noise-bandwidth/.
  Contains nice definition, and graphics
- MATLAB source for simulation: https://fr.mathworks.com/help/signal/ref/enbw.html#btrihsq-1

ENBW

ENBW: To redo a proper documentation and simulation from the resources above.

Chapter 7

FIR

7.1 Window Method

7.1.1 Low Pass Design

In this example, we design a low pass FIR filter using 2 windows: rectangular and Hamming.

The rectangular provides the best transition bandwidth, but has unacceptable ripples in the pass band and the stop band, so we use Hamming window to adjust these unacceptable ripples in the expense of a wider transition bandwidth.

The design specifications are:

- Order: 24
- Normalized cuttoff frequency: 0.5

Results are in Figure 7.1.

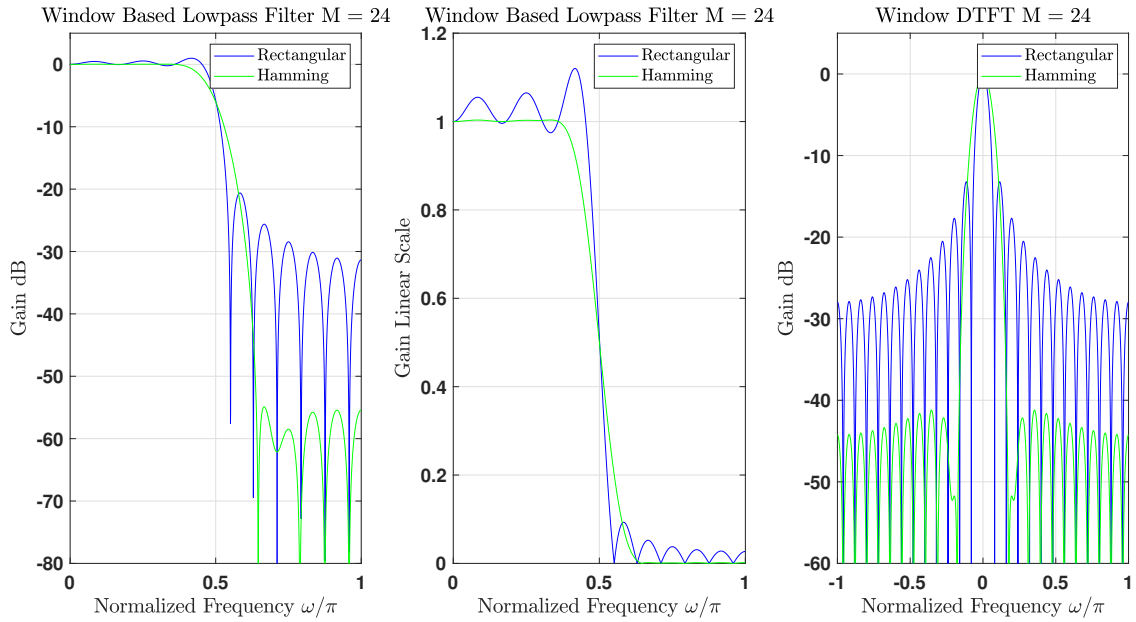


Figure 7.1: FIR Low pass example using Rectangular and Hamming window

- For Hamming window:
  - Main lobe width is $\frac{8\pi}{M}$ where M is the filter order in this case
  - Peak side lobe is -41 dB

7.1.2 Bandpass Design

Now we will do a banpass design using the following specs shown in Figure 7.2



Figure 7.2: Bandpass Design Example

- In the 1st stop band, the transition bandwidth is $\frac{2\pi}{5} - \frac{\pi}{5} = \frac{\pi}{5}$
- In the 2nd stop band, the transition bandwidth is $\frac{7\pi}{10} - \frac{3\pi}{5} = \frac{\pi}{10}$
- So as a result and specification we choose the smaller transition bandwidth that is $\frac{\pi}{10}$

- As a deviation error, we have $0.01 < 0.05$, so we take 0.01 as the error to be computed in the stopband and the bandpass

The window to be chosen for the specification is the Kaiser window, which depends upon 2 parameters:

1. β which controls the window shape: if $\beta = 0 \rightarrow$ rectangular window, and if β increase then the mainlobe width increase and the sidelobe height decrease
2. M for the transition bandwidth

Results are shown in Figure 7.3 and Figure 7.4.

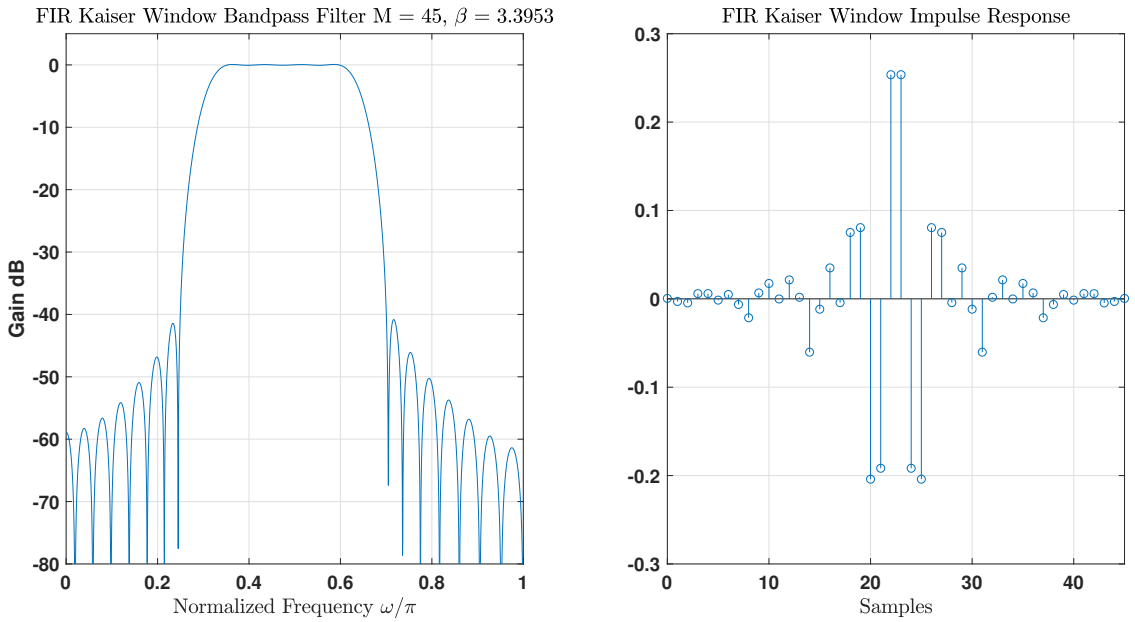


Figure 7.3: Bandpass Design Results for Kaiser Window

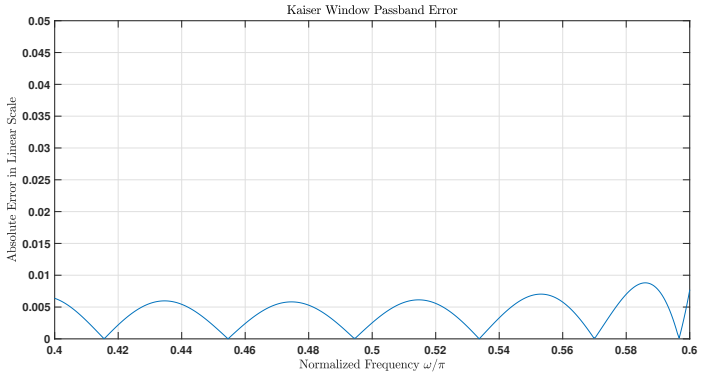


Figure 7.4: Bandpass Design Results for Kaiser Window: Error is less then 0.01

7.2 Frequency Sampling Method

7.2.1 Low Pass Example

The specifications are:

- $\omega_{\text{pass}} = 0.25\pi$, $\omega_{\text{stop}} = 0.35\pi$
- $A_p = 0.1$ dB , $A_s = 40$ dB
- By result the transition bandwidth is $\Delta\omega = \omega_{\text{pass}} - \omega_{\text{stop}} = 0.35\pi - 0.25\pi = 0.1\pi$

For a rectangular window as interpolating function, we have $\Delta\omega = \frac{4\pi}{L} \rightarrow L = 40 = M + 1$, so we need to begin with an ***FIR of length at least 40*** (order 39).

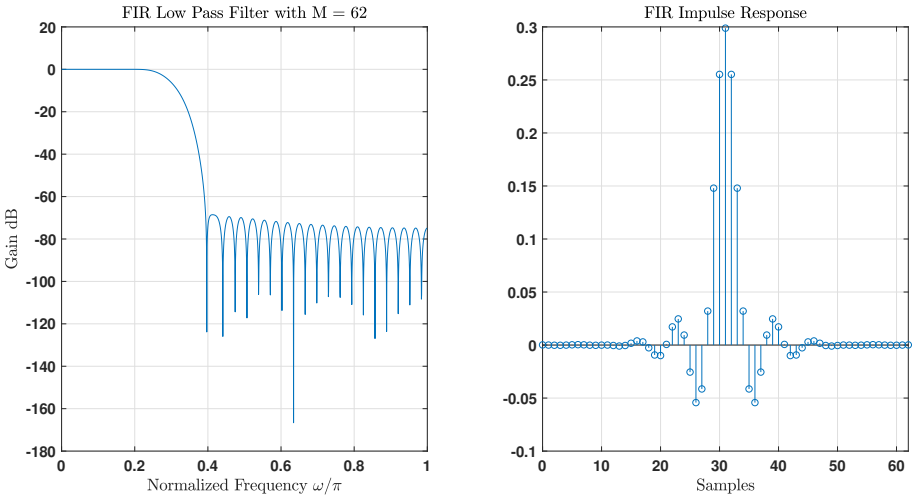


Figure 7.5: FIR Low Pass with an order of $M = 62$

To do

:

- Write the formula and the design criteria
  - Eq of FIR
  - Steps of design in window method
  - Concepts and techniques in designing
  - Implementation using toolbox and from scratch
  - Repeat the examples using toolbox and implementation using OOP style

Chapter 8

Matched Filter

8.1 Intro and Recap

Before simulation we will do a recap about matched filter principles (oriented in communication application).

Imagine we have some bits to be transmitted (1 is encoded as 1 volt and 0 as -1 volt), where these bits are passed through a pulse shaping filter (we will assume we have a square pulse wave for a period of T_{bit}).

Block diagra

Block diagram: To insert a block diagram of the intuition later.

These pulses will pass through a channel, and will have some gain G and some additive noise $n(t)$.

Question: How do we know if we receive a +1 or -1 ?

Answer: Well, we have to collect all the energy through the pulse, that's why we need to design a filter $h(t)$ which do this, and *then* sample its output at multiple time of $T_{\text{bit}} \leftrightarrow t = K\,T_{\text{bit}}$.

The following fact (without the proof) are to be remembered:

- The best $h(t)$ is to be known $s(T_{\text{bit}} - t)$: that is the same pulse shape, but flipped in time and then time shifted by T_{bit}
  - As a matter of fact, the formal definition of a matched filter: it is a filter designed to maximize the SNR $\rightarrow$ the SNR will be maximized at the communication (or any system we use it)
- For a square pulse $s(t)$ for example, $h(t)$ will be also a square wave for this particular case. The result of a convolution of 2 square wave will be a triangular wave of period $2\,T_{\text{bit}}$.
- for other pulses, we will have also triangles but shifted

- At each peak time, we will not have a contribution from other triangles → which means that we don't have contribution from other pulses, and only current pulse is contributing

Chapter 9

Bandpass Sampling

9.1 Non Integer Band Positioning

9.1.1 Reminder

Source: see section 2-3 in [11].

- In bandpass sampling, we are interested in bandwidth rather the highest frequency component in the spectrum
- Sampling rate bounds to respect [11]:

$$\frac{2\,f_{\text{center}} + B}{m} \leq F_s \leq \frac{2\,f_{\text{center}} - B}{m - 1} \tag{9.1}$$

  such that $m \in \mathbb{N}$ represent the number of replica.

9.1.2 Simulation

Chapter 10

Introduciton to Multirate Signal Processing

10.1 Donwsampling

What happen to the time frequency index when doing downsampling.

From frequency sampling and DFT computations, we know that when computing a frequency bin numerically, the frequency step is $\frac{2\pi}{N}$, where N is the total number of samples (the length of the discrete time signal or the sampled signal).

So we have the relation:

$$f \times t \leftrightarrow n \times \frac{k}{N} \tag{10.1}$$

where:

- f and t are some indexes for continuous frequency and time
- n: discrete time index
- k: discrete frequency index

When we downsample by a factor of D, the relation between the indices is:

$$\frac{k\,n}{N} \leftrightarrow \frac{k\,n}{N/D} \tag{10.2}$$

Also recall that $F_s \geq 2\,f$, but a more equivalent common form is $f \leq \frac{F_s}{2}$, which mean the analog baseband frequency is captures between the range 0 to $\frac{F_s}{2}$. So we can say:

$$F_s \geq 2\,f \equiv f \leq \frac{F_s}{2} \tag{10.3}$$

What happen when we downsample?

If we begin with a signal with a sampling rate F_s, where the CTFT spectrum contains N point (N is the total number of samples with a sampling period $T_s = \frac{1}{F_s}$).

Let's take an example of a sin wave, where its CTFT spectrum is shown in Figure 10.1.

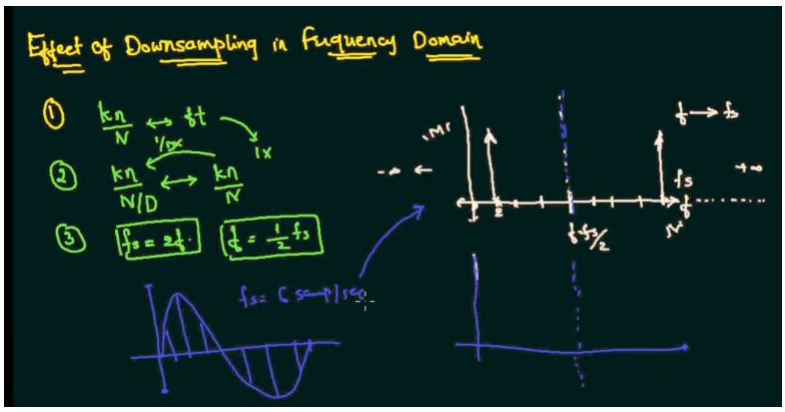


Figure 10.1: Sine Wave CTFT Spectrum

The original $F_s = 6Hz$, that is 6 sample per cycle (this in the time domain), and the DFT spectrum for the CTFT contains $N = 6$ frequency fundamental points (for a window duration of 1s let's say).

If we downsample and let the new F_s 3 Hz, that is $F_s^D = 3$ Hz, we will have:

- 3 samples per cycle (period) instead of 3 $\leftrightarrow$ time domain effect
- Frequency domain effect (see Figure 10.2):
  - Folding line $\frac{F_s}{2}$ will get closer
  - Frequency points indices will also get lost because N is reduced , but the frequency index 2 remain fix $\leftrightarrow$ so the spectrum get preserved

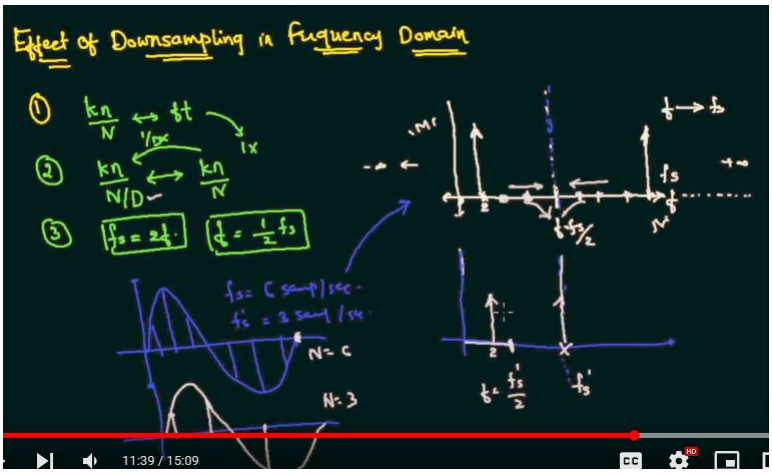


Figure 10.2: Sine Wave CTFT Spectrum

Important note: if we do dowsanmpling, it is not necessarily that aliaising get occurred.

A more general example for a bandlimited signal is shown in Figure 10.3



Figure 10.3: Sine Wave CTFT Spectrum

When we downsample, the folding line $\frac{F_s}{2}$ will get closer (move to the left). As long as the new folding line $\frac{F_s^D}{2} > f_{max}$, the signal spectrum is preserved and no aliasing occur.

10.1.1 Example: Downsampling with no aliasing

dowsamplin with no alia ing

: to redo with a sinc example.
We take a numerical example to illustrate downsampling.

First we generate a bandlimited signal with a bandwidth of 100 Hz, generated using the `fir2` command.

The $\omega_{\text{pass}} = 0.25\pi$, and the correspondent analog frequency $f_{\text{pass}} = \omega_{\text{pass}} \times F_s \times \frac{1}{2\pi} = 100Hz$.

The initial sampling rate $F_s^{\text{init}} = 800Hz$, and we downsample by a factor 4, so the new sampling rate $F_s^D = \dfrac{F_s^{\text{init}}}{D} = \dfrac{800}{4} = 200$ Hz.

Notice that new folding line now is $\dfrac{F_s^D}{2} = 200 > 100$, where the 100 is the bandwidth of the signal, so no aliasing will occur and the bandwidth will still the same.

10.2 Downsampling with Aliasing

Note: this section contains many simulations and explanation, and since they have allot of details and contain many points, I decided to make it as a section instead of a subsection.

10.2.1 Example 1: Aliased Sin and Cos

Ref: chapter 11 in [12].

We take 1 cos and sin signal as the following:

$$x(t) = 5\sin(2\pi 10^3 t) + \cos(2\pi\, 2.5 \times 10^3 t) \tag{10.4}$$

- Frequency of each component is at 1 KHz and 2.5 KHz
- $F_s = 8$ KHz

For $F_s = 8$ KHz, the folding frequency is at $\frac{F_s}{2} = 4$ KHz > 2.5KHz, so no aliasing occur.

Let's say we downsample by a factor $D = 2$, so we will have:

- $F_{s_D} = \frac{F_s^{\text{orig}}}{D} = \frac{8000}{2} = 4$ KHz
- The new folding frequency will be $\frac{4000}{2} = 2000$ Hz < 2500 Hz

In this case, the new folding frequency is smaller then the bigger frequency component in $x(t)$, so aliasing will occur in the reconstructed signal.

Computation of the aliased frequency:

The alias frequency will be smaller then the new sampling rate $F_{sD} = 4$ KHz, and it can be computed as:

$$\begin{aligned} f_{\text{alias}} &= F_{sD} - f_{\max} \\ &= 4 \text{ KHz} - 2.5 \text{ KHz} \\ &= 1.5 \text{ KHz} \end{aligned} \tag{10.5}$$

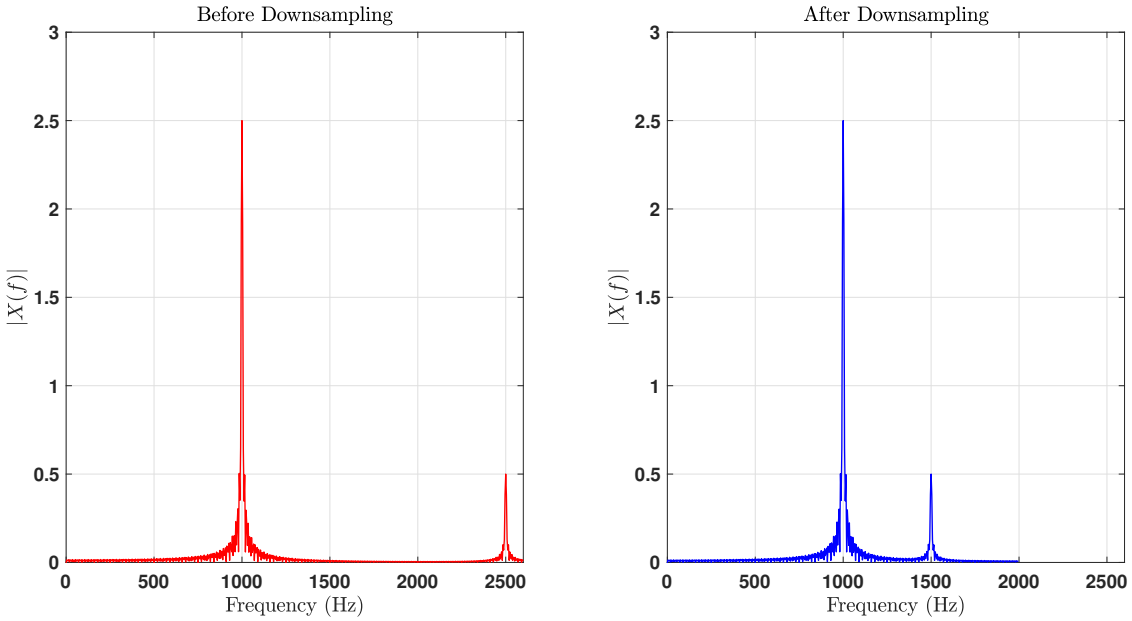


Figure 10.4: Sine Wave CTFT Spectrum

Example 2:

In example 1, we have illustrated the aliasing by keeping the frequency of the signal constant and changing F_s, but also we can illustrate the aliasing by changing the frequency of the signal (like we have some signal generator with variable input frequencies) and keep F_s constant.

10.2.2 Example 3: Aliasing with a full bandlimited signal

Example 1 in 10.2.1 focus on visualizing the aliasing with only sin (or cos) signals, or the sum of many sin.

The sin spectrum consist only of 1 peak at the location of frequency of the sin. Now we try to simulate the aliasing with signals where the spectrum extend from $0 \rightarrow B$[Hz].

Note of vocabulary: by *full bandlimited signal*, I mean that the amplitude spectrum is present over all the bandwidth B, not only over 1 frequency component like the sin or cos signals
So as example, we take the sinc signal.

Flow of the explanation

- When I was try to debug the problem, I deiced to start with other test signal then sin or cos $\leftrightarrow$ signals that have only 1 frequency components
- As a choice I took a sinc signal (See section 3.2.3 in [4]):

$$\text{sinc}\left(\frac{t}{T}\right) \leftrightarrow X(\Omega) = T \quad \text{for } \frac{-\pi}{T} \leq \Omega \leq \frac{\pi}{T} \tag{10.6}$$

- Parameters in simulation:
  - $B = 10$ Hz
  - A *big time window* from $-50 \rightarrow 50$ s
    - So when simulating aliasing phenomena, I ensure that my source of error is only in low F_s and not in a small time window
    - Also I start before $t = 0$, so I don't assume causality of signal for now
- Result of Input Analog Signal:

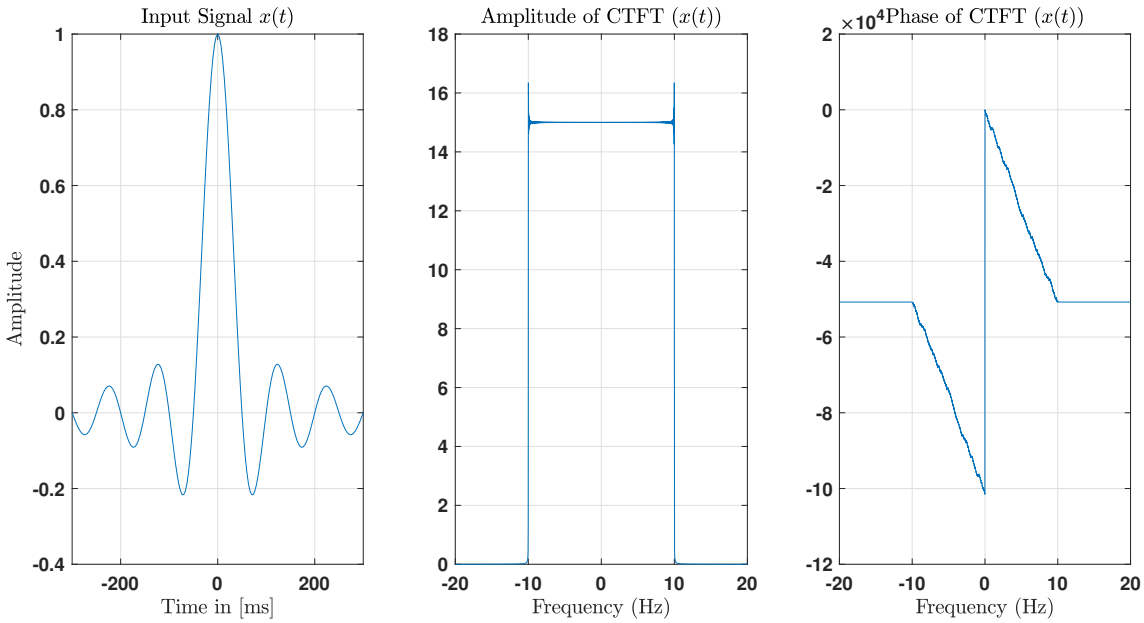


Figure 10.5: Input Analog Sinc signal

  - For analog simulation, I use oversampling to create a fine grid of analog time where $F_{s_{\text{analog}}} = \text{oversampling} \times B$.
    In the code I took the oversampling $= 30 \rightarrow F_{s_{\text{analog}}} = 300$ Hz
  - For now I don't use any postprocessing (divide by the number of samples for example in a sin or a cos signal) for the FFT command to adjust the magnitude or the phase.
    I focus for now that the shape of the magnitude spectrum is correct with a correct bandwidth

- To simulate any other F_s, I use *integer* dowsampling operation where $D = \frac{F_{s_{\text{original}}}}{F_{\text{starget}}} \in \mathbb{N}$ such that $F_{\text{starget}} < F_{s_{\text{original}}}$
- To ensure my dowsampling operation is correct, I choose 2 $F_{\text{starget}} = [60\,,\,30]$ Hz
  - Both $F_{\text{starget}} > F_{\text{Nyquist}} = 2 \times F_H = 20$ Hz $\rightarrow$ no aliasing occur, and we will be able to do reconstruction

Results: CTFT Amplitude Spectrum

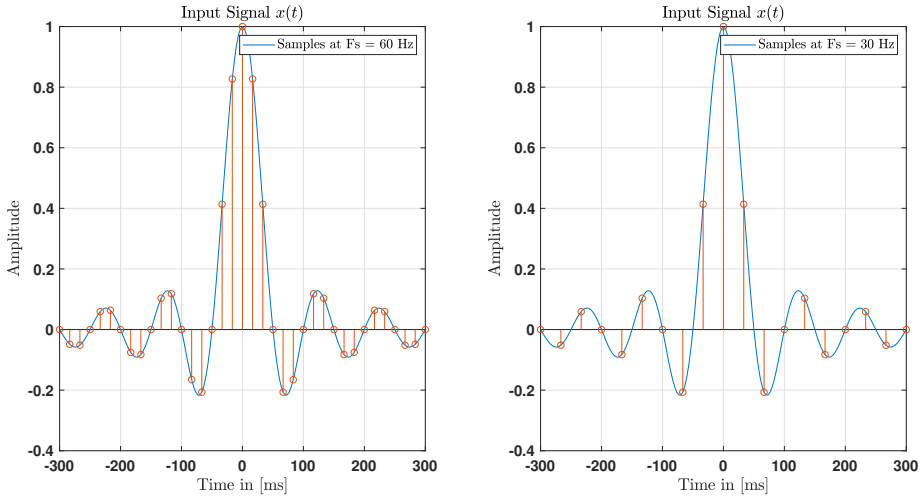


Figure 10.6: Ideal Reconstruction

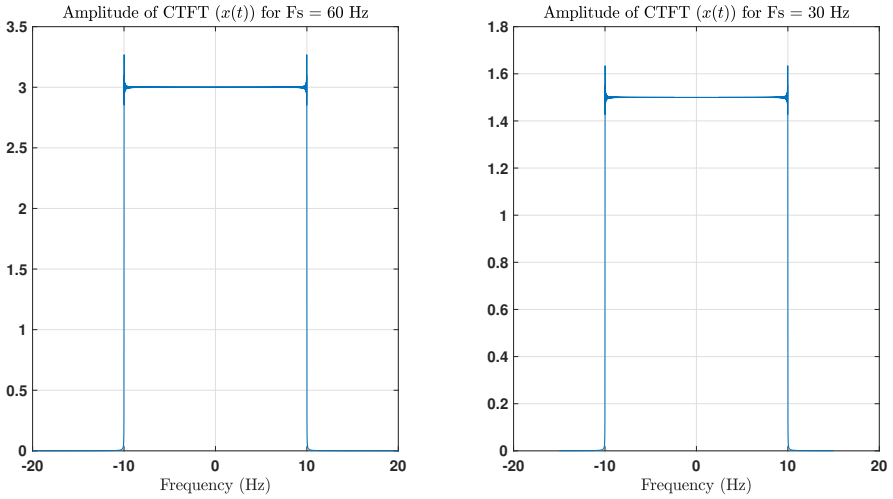


Figure 10.7: CTFT Amplitude Spectrum

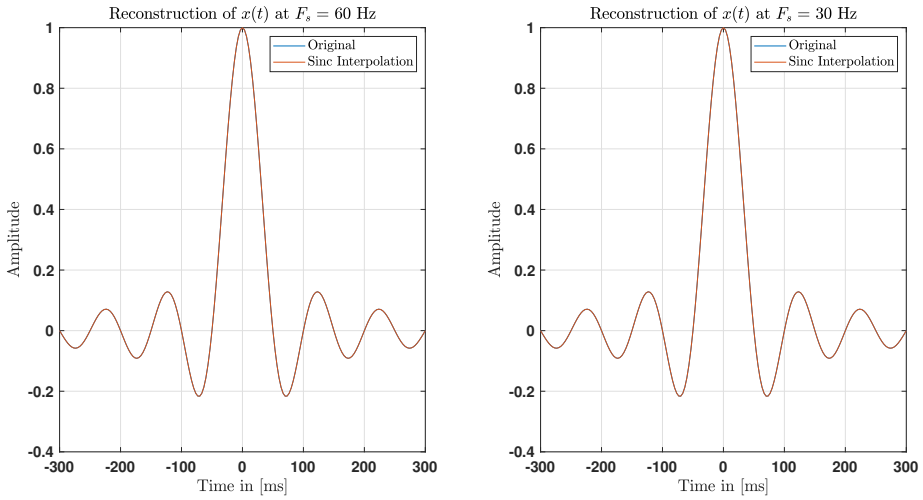


Figure 10.8: Ideal Reconstruction

- Now I decide to choose a $F_s < F_{\text{Nyquist}}$
  - From theory, when we drop F_s below $F_{\text{Nyquist}} \equiv \Omega_s < 2\,\Omega_H$, *aliasing* should occur

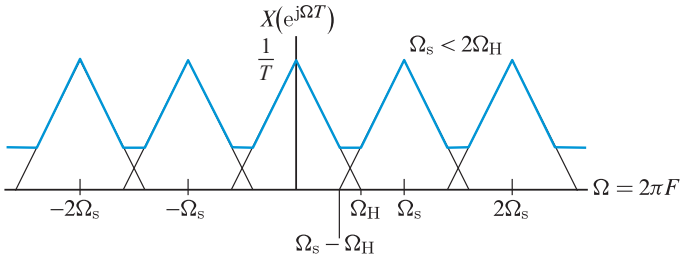


Figure 10.9: Aliasing when $\Omega_s < 2\,\Omega_H$ [1]

  - From Figure 10.9, we can say that:
    1. Part of the spectrum is preserved and the other part which *don't respect Nyquist theorem* is distorted
    2. But the signal spectrum still at Ω_H but distorted (I will clarify this point later in simulation)
  - I tried to take $F_{s_{\text{ch}}} = 15$ Hz
  - Result:

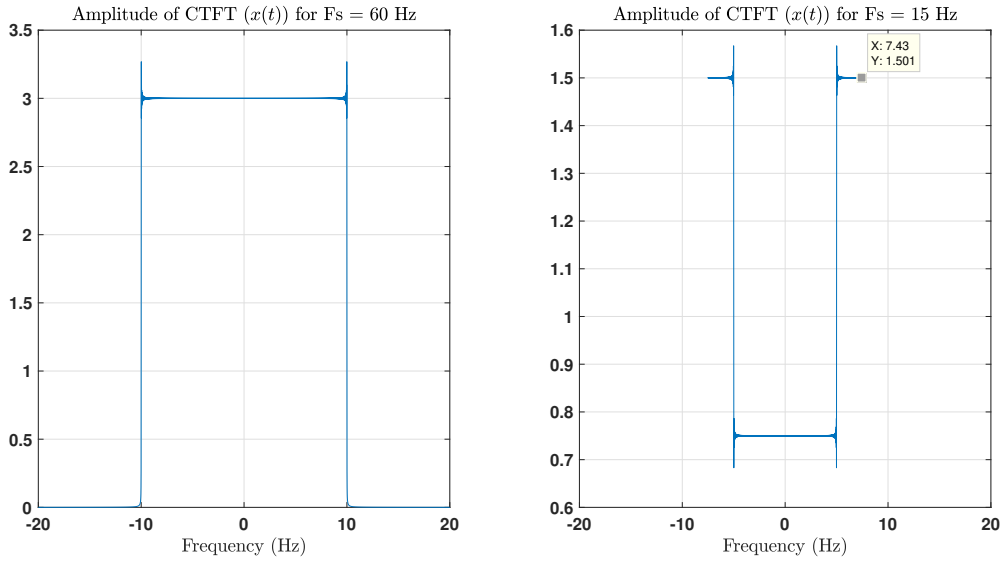


Figure 10.10: Aliasing using $F_{s_{\text{ch}}} = 15$ Hz

  - the $F_s = 60$ Hz is just a reference, to compare aliased and unaliased spectrum
  - For $F_s = 15$ Hz, my spectrum limit stop at 7.5 Hz and not at $B = 10$ Hz
    * From my understanding of aliasing, I should get a spectrum of width $= B = 10$ Hz, but distorted some how (addition of replica,from frequency exactly start the distortion,$\cdots$)

* This is because usually when simulating aliasing , I use fft(samples), where the samples are the downsampled giving a rate of 15 Hz (this is logic)
* *But* (this is the important part) for frequency bins (which we as programmers generate): I generate it from $\frac{-F_s}{2} \rightarrow \frac{F_s}{2}$
* Doing so is fine when $F_s > F_{\text{Nyquist}}$: In the left plot of Figure 10.10 for $F_s = 60$ Hz, I did so and the spectrum is fine
* For the case of aliasing where $F_s = 15$ Hz, $\frac{-F_s}{2} \rightarrow \frac{F_s}{2} = -7.5 \rightarrow 7.5$ Hz
* However, I didn't in my life generate a frequency bins when taking an alias case, since in normal signal processing, we prevent aliasing and respect Nyquist sampling condition
* Clearly I need a more general way to implement aliasing in a correct way

- I did some search, and I found this example from matlab documentation
  - As motivation, when I search for examples to simulate aliasing over a full range of frequencies from $0 \rightarrow B$[Hz], it was the only convenient one
  - Put the picture of the replicated spectrum when doing sampling, and say that my target is to simulate the idea that when sampling below Nyquist rate, distortion arise
  - All other resources focus only on sin or cos (so only 1 active frequency component).

    aliasing resources

    to cite the aliasing resource I have

- The example use discrete time approach, which is shown in Figure 10.11

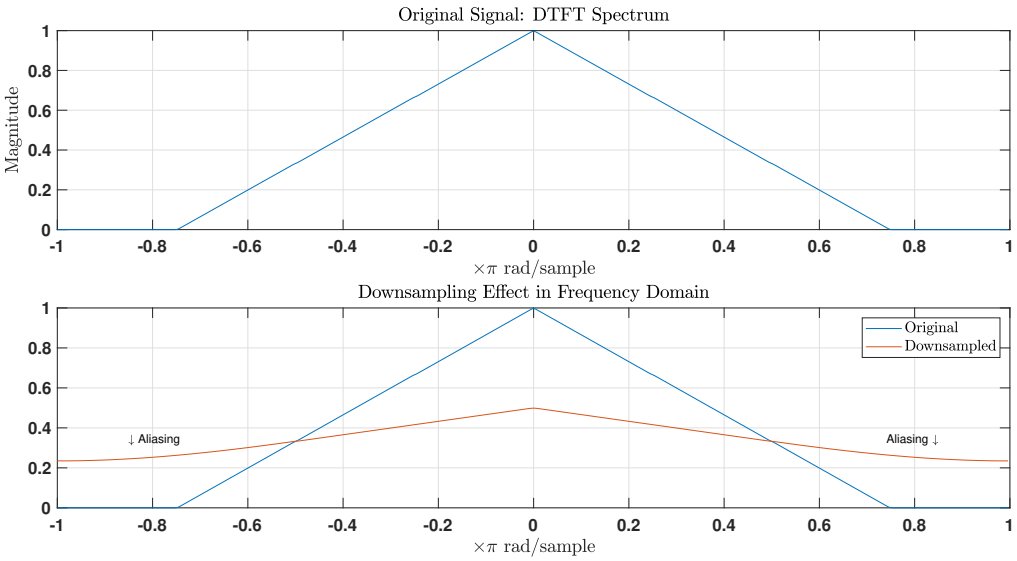


Figure 10.11: Aliasing Example in discrete approach

- Explanation of Figure 10.11
  - A *discrete time bandlimited signal* is generated, where the *discrete bandwidth* $= 0.75\pi$ rad/cycle
  - We do an integer downsampling operation of $D = 2$
  - Recall that dowsampling doesn't necessary causes aliasing
  - To know whether aliasing occurred or not, we test the following condition:

$$\text{Discrete Spectral support} < \frac{2\pi}{D} \tag{10.7}$$

    * If this is true, then dowsampling by $D \in \mathbb{N}$ won't cause aliasing
    * If not, then aliasing occur
  - Back to the example, the *Discrete Spectral support* contains *positive discrete bandwidth* and the *negative part* $\rightarrow$ *Discrete Spectral support* $= 0.75\pi \times 2 = 1.5\,\pi$ rad/cycle

  - Testing Equation 10.7 for $D = 2$, we get: $1.75\pi > \dfrac{2\pi}{D=2} = \pi \rightarrow$ aliasing occur
    * Spectral copies will overlap and get added
  - Although that aliasing is shown, *but* there is no mention to any F_s or (T_s): the example is simulated assuming that we are getting the discrete samples some how
    * Recall that in our case (not just my case but also in real world), we express everything in Hz, seconds,$\cdots \leftrightarrow$ some meaningful units
    * To get back from from radian/cycle $\rightarrow$ Hz, we use [1]:

$$F\,[\text{Hz}] = \omega[\text{radian/cycle}] \times \left(\frac{F_s}{2\pi}\right) \tag{10.8}$$

    * However, Equation 10.8 is normally used when Nyquist theorem is respected $\leftrightarrow F_s > 2 \times f_{\max}$
    * But in our case, it is not
    * So this give me a question, what F_s should I use to obtain correctly a distorted spectrum according to Figure 10.9
  - Put frequency conversion formula
  - All the overlapping and the spectrum is inside $[-\pi\,,\,\pi]$, even when aliasing occurs
  - Show the plot of a aliased sinc now, and compare it to the left plot of Figure 10.10

- Applying concept in the example of Figure 10.11 to my case of sinc signal, where $B = 10$ Hz and $F_{s_{\text{ch}}} = 15$ Hz:
- Now when finishing from simulating in the discrete frequency domain, we need to move to the continuous frequency domain (that is F[Hz])
  - However, I have a problem which F_s to use when doing the conversion

Chapter 11

Random Process

In this chapter, we see how to generate white noise with correct PSD.

Assuming a baseband model of bandwidth B Hz, and AWGN channel, the total noise power is the area

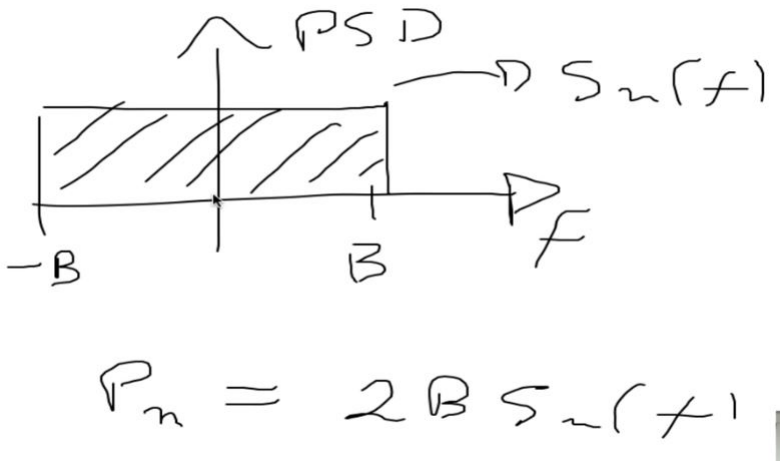


Figure 11.1: PSD for AWGN Baseband Model

From random process, the power of random sequence $X[n]$ is equal to the mean square, that is $E[X^2]$.
Now if this sequence have zero mean, then the mean square will be equal to the variance $\leftrightarrow E[X^2] = \sigma$.
So if want to generate some noisy sequence having some total power (whih is equal to σ), we simply multiply the noisy sequence generated with the standard deviation $\sqrt{\sigma} \leftrightarrow$

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